



CertusPro-NX LPDDR4 Controller IP Core

User Guide

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Contents

Acronyms in This Document	7
1. Introduction	8
1.1. Quick Facts	8
1.2. Features	8
2. Functional Descriptions	10
2.1. Overview	10
2.2. LPDDR4 PHY Module	10
2.2.1. Initialization and ZQ Calibration	10
2.2.2. Command Bus Training (CBT)	10
2.2.3. Write Leveling	11
2.2.4. Read Preamble Training	11
2.2.5. Read Training.....	11
2.2.6. Write Training.....	11
2.2.7. Error Code and debug	12
2.3. LPDDR4 Controller Module	13
2.3.1. Initialization	13
2.3.2. Refresh and Self-Refresh Management	14
2.3.3. Local-to-Memory Address Mapping.....	14
2.3.4. Burst and Address Management.....	14
2.3.5. Local Bus Timing	15
2.3.6. Write Operation	16
2.3.7. Mask Write Operation.....	17
2.3.8. Read Operation	17
3. IP Parameter Description.....	19
4. Signal Descriptions.....	23
5. Registers Description	26
5.1. Feature Control Register 0	26
5.2. Feature Control Register 1	27
5.3. Status Register 0.....	29
5.4. Status Register 1.....	30
5.5. Status Register 2.....	30
5.6. Interrupt Register	31
6. IP Core Generation and Evaluation	33
6.1. Mask Write Function Setting.....	35
6.2. Self-Refresh Power Down Function Setting	36
6.3. Timing Constraint.....	36
7. Example Design and Evaluation.....	38
7.1. Simulation	39
7.2. Evaluation Design.....	40
7.3. Evaluation Tool.....	40
8. User Interface bridge reference design	45
8.1. Virtual FIFO.....	45
8.1.1. Read Data Latency.....	45
8.1.2. Outstanding Command Counters.....	46
8.2. AXI-4 I/F.....	50
8.3. Avalon I/F	55
8.4. Multi-Port Memory Controller reference design	59
9. Signal Integrity and PCB Guidelines.....	64
9.1. Topology.....	64
9.1.1. T-Balance	64
9.1.2. Flyby	64
9.2. Reference Clock.....	65

9.3. Drive Strength and Termination.....65

9.4. CertusPro-NX LPDDR4-1066 Interface Layout Example65

References71

Technical Support Assistance72

Revision History73

Figures

Figure 2.1 LPDDR4 SDRAM Controller Block Diagram	10
Figure 2.2 Initialization success	13
Figure 2.3 Initialization fail	13
Figure 2.4 Self-Refresh Enter and Exit	14
Figure 2.5 Local-to-Memory Address Mapping for Memory Access	14
Figure 2.6 Local Bus Command Port	15
Figure 2.7 Local Bus Data Port	16
Figure 2.8 Write Operation	17
Figure 2.9 Mask Write Operation	17
Figure 2.10 Read Operation	18
Figure 6.1 GUI Controller Settings	33
Figure 6.2 GUI PHY Setting	34
Figure 6.3 GUI Memory Setting	35
Figure 6.4 Self-Refresh Power Down Function Setting	36
Figure 6.5 Directory of Constraint File	36
Figure 6.6 Post-Synthesis Timing Constraint Editor	37
Figure 6.7 Setting of Constraint	37
Figure 7.1 LPDDR4 Core Directory Structure	38
Figure 7.2 Normal Simulation Configuration	39
Figure 7.3 Accelerated Simulation Configuration	40
Figure 7.4 Auto Test Mode	41
Figure 7.5 Manual Test Mode	42
Figure 7.6 DevTools	43
Figure 7.7 CMD Analyze	44
Figure 8.1 Virtual FIFO Diagram	45
Figure 8.2 Read Timing with Output Register Bypassed	45
Figure 8.3 Read Timing with Output Register Enabled	46
Figure 8.4 Outstanding Write and Read Command Counter Timing	46
Figure 8.5 Virtual FIFO Configuration GUI	47
Figure 8.6 AXI-4 Configuration GUI	51
Figure 8.7 Avalon Configuration GUI	56
Figure 8.8 MPMC Typical usage	60
Figure 8.9 MPMC Cascade	60
Figure 8.10 MPMC IP GUI	61
Figure 9.1 T-Balance Topology	64
Figure 9.2 Flyby Topology	64
Figure 9.3 Lattice LPDDR4 controller subsystem on PCB	66
Figure 9.4 Reference PCB Stack information	66
Figure 9.5 LPDDR4 DQS group A routing Example	66
Figure 9.6 LPDDR4 DQS group B routing Example	68
Figure 9.7 LPDDR4 CA group routing Example with Dual Rank support	69
Figure 9.8 LPDDR4 CA group split Example with T-type topology	69

Tables

Table 1.1 CPNX LPDDR4 Controller IP Quick Fact	8
Table 2.1 Error Code Description	12
Table 3.1 Controller Attributes Setting	19
Table 3.2 Memory Attributes Setting	21
Table 3.3 PHY Attributes Setting	21
Table 4.1 Port Definitions	23
Table 5.1 Register Map	26
Table 5.2 Feature Control Register 0 Information(user_apb_paddr_i=00H).....	26
Table 5.3 Feature Control Register 1 Information(user_apb_paddr_i=04H).....	27
Table 5.4 Status Register 0 Information(user_apb_paddr_i=08H)	29
Table 5.5 Status Register 1 Information(user_apb_paddr_i=0CH)	30
Table 5.6 Status Register 2 Information(user_apb_paddr_i=10H)	30
Table 5.7 Interrupt Register Information(user_apb_paddr_i=88H)	31
Table 6.1 Truth Table of IP configuration and related mask operation	35
Table 7.1 Test Mode Select	41
Table 8.1 Virtual FIFO Design Quick Fact	47
Table 8.2 Virtual FIFO Design Ports	47
Table 8.3 Virtual FIFO Design Parameters	49
Table 8.4 AXI I/F Resource	51
Table 8.5 AXI I/F Performance	51
Table 8.6 The port of AXI-4 I/F	53
Table 8.7 The parameters of AIX-4 I/F	54
Table 8.8 The address width setting for LocalBus Address Width and AXI Address Width	54
Table 8.9 The AxLEN setting	55
Table 8.10 Resource of Avalon I/F	56
Table 8.11 Performance of Avalon I/F	56
Table 8.12 The Interfaces Description of Avalon to LocalBus	58
Table 8.13 The Parameter Description of Avalon to LocalBus	59
Table 8.14 The address width setting for LocalBus Address Width and Avalon Address Width	59
Table 8.15 The Avalon burst count setting	59
Table 8.16 MPMC Resource	61
Table 8.17 Order Fixed Priority	61
Table 8.18 Round Robin Priority	62
Table 8.19 MPMC Parameters	62
Table 8.20 MPMC Ports	62
Table 9.1 FPGA Side Settings	65
Table 9.2 LPDDR4 Side Settings	65
Table 9.3 PCB Layout general guidance	65
Table 9.4 DQS group A trace information of example PCB	67
Table 9.5 DQS group B trace information of example PCB	68
Table 9.6 CA group trace information of example PCB	69
Table 9.7 LPDDR4 IP configuration parameters based on example PCB	70

Acronyms in This Document

A list of acronyms used in this document.

Acronym	Definition
CA	Command & Address
CBT	Command Bus Training
CPNX	CertusPro-NX
DDR	Double Data Rate
DQS	Data Strobe
FIFO	First In First Out
FPGA	Field Programmable Gate Array
GUI	Graphical User Interface
I/O	Input/Output
IP	Intellectual Property
LPDDR	Low Power Double Data Rate
MC	Memory Controller
MPMC	Multi-Port Memory Controller
SDR	Single Data Rate
SDRAM	Synchronous Dynamic Random-Access Memory
VREF	Voltage Reference

1. Introduction

The Low Power Double Data Rate (LPDDR4) Synchronous Dynamic Random Access Memory (SDRAM) Controller is a general-purpose memory controller that interfaces with standard LPDDR4 memory devices/modules compliant with JESD209-4, LPDDR4 SDRAM Standard, and provides a generic command interface to user applications.

The CertusPro-NX LPDDR4 Controller IP consists of a controller, DDR PHY, and associated clocking and calibration logic to interface with LPDDR4 SDRAM.

1.1. Quick Facts

Table 1.1 provides quick facts about CertusPro-NX LPDDR4 SDRAM Controller IP.

Table 1.1 CPNX LPDDR4 Controller IP Quick Fact

Support Device	CertusPro-NX Family					
Resource Utilization (Single Rank)	Data Width	16bit	24bit	32bit	48bit	64bit
	LUTs	3327	3591	3812	4374	5326
	Block RAM	18EBR /2EBR+1LRAM	18EBR /2EBR+1LRAM	18EBR /2EBR+1LRAM	18EBR /2EBR+1LRAM	18EBR /2EBR+1LRAM
Resource Utilization (Dual Rank)	Data Width	16bit	24bit	32bit	48bit	64bit
	LUTs	3465	3696	3937	Not Support	
	Block RAM	34EBR /2EBR+1LRAM	34EBR /2EBR+1LRAM	34EBR /2EBR+1LRAM		
Performance	Burst Count	1	2	4	6	8
	Efficiency (W/R mix)	21%	29%	40%	48%	54%
	Burst Count	10	12	16	32	64
	Efficiency (W/R mix)	58%	62%	68%	79%	87%
	Read Latency	16 sclk				
	Write Latency	5 sclk				

1.2. Features

The CertusPro-NX LPDDR4 Memory Controller supports the following key features:

- Support for CertusPro-NX devices.
- Interfaces to industry standard LPDDR4 SDRAM components and modules compliant with JESD209-4, LPDDR4 SDRAM Standard
- Extreme low read latency, 64tCK (16sclk)
- LPDDR4 SDRAM speeds of
 - 266MHz (533Mbps)
 - 400MHz (800Mbps)

- 500MHz (1000Mbps)
- 533MHz (1066Mbps)
- 600MHz (1200Mbps) used for stress test
- 700MHz (1400Mbps) used for stress test
- 750MHz (1500Mbps) used for stress test
- 800MHz (1600Mbps) used for stress test
- Memory data path widths of x16, x24, x32, x48, x64 bits
- For CertusPro-NX 100K
 - Maximum number of instantiations: 2
 - Maximum total DQ width: 80 bit(48bit+32bit)
- Burst Count 1~64 (64*BL16 can cover one row)
- Write Mask function
- Self-Refresh and Power Down function
- Automatic LPDDR4 SDRAM initialization and refresh
 - Automatic CBT calibration including Vref (CA) and de-skew.
 - Automatic Write Leveling
 - Automatic Read Preamble training
 - Automatic Read/Write training including Vref (DQ) and de-skew.
- Support On-Die Termination (ODT)
- Support Single-Rank and Dual-Rank package memory

2. Functional Descriptions

2.1. Overview

The LPDDR4 SDRAM controller IP consists of two major parts: Controller module and PHY module. The user interface of IP is Local Bus, detail please refer [Local Bus Timing](#). MPMC, Video Frame Buffer, Cache, AXI4 interface, Avalon interface, Virtual FIFO interface and so on are provided as other independent IP which can connect to Local bus. UART and APB BUS are used for configuration and debugging, there are not necessary interfaces for user. [Figure 2.1](#) provides a high-level block diagram illustrating the main functional blocks and the technology used to implement the LPDDR4 SDRAM Controller IP core functions.

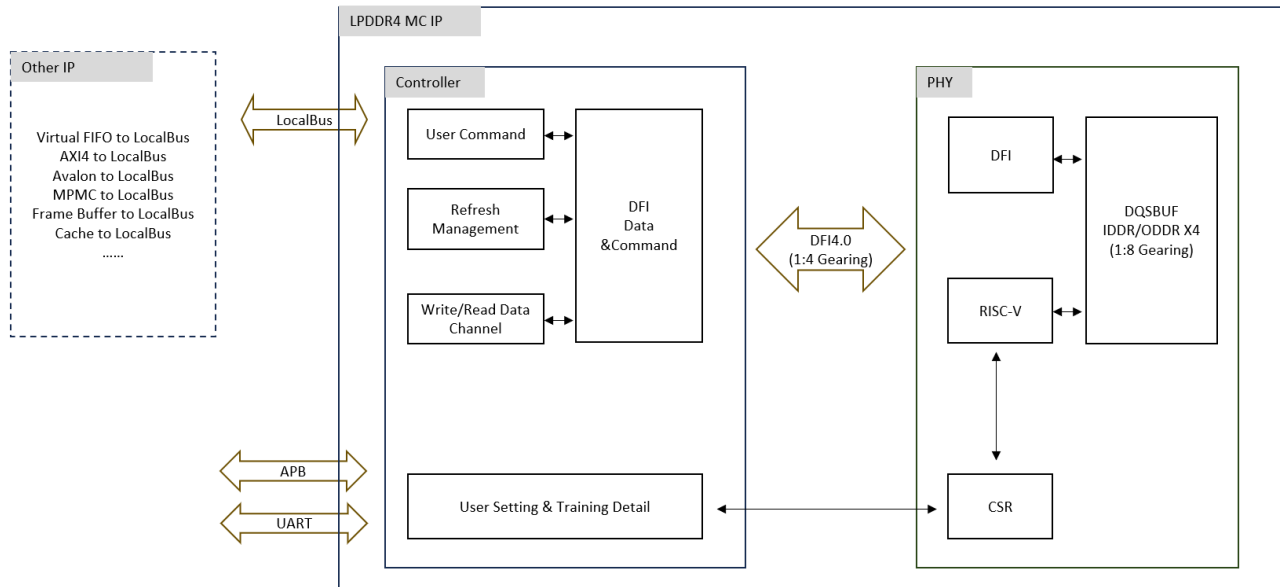


Figure 2.1 LPDDR4 SDRAM Controller Block Diagram

2.2. LPDDR4 PHY Module

The LPDDR4 PHY module implements soft logic in the FPGA fabric for Initialization, CBT training, Write leveling, Read preamble training and Read data training, Write data training. There is also hard block DQSBUF and Delay unit used to support 1:4 clock gearing (1:8 data gearing) VREF generate and adjust IO delay.

2.2.1. Initialization and ZQ Calibration

The Initialization block performs the LPDDR4 memory initialization sequence as defined by JEDEC protocol. After power on or a normal reset of the LPDDR4 controller, memory must be initialized before sending any command to the controller. User needs to assert the `phy_init_start_i` input to the LPDDR4 controller to start the memory initialization sequence. The completion of initialization is indicated by the `phy_init_success_o` or `phy_init_fail_o` output provided by this block.

2.2.2. Command Bus Training (CBT)

The LPDDR4-SDRAM command bus must be trained before enabling termination for high-frequency operation. LPDDR4 provides an internal Vref (CA) that defaults to a level suitable for un-terminated, low frequency operation, but the Vref (CA) must be trained to achieve suitable receiver voltage margin for terminated, high-frequency operation. The training methodology centers the Vref (CA) in the CA data eye and at the same time allows for timing adjustments of the CS and CA signals to meet setup/hold requirements.

2.2.3. Write Leveling

LPDDR4 memory module used fly-by topology for the CA Bus and CK signals for better signal integrity. This reduces the number of stubs and length, but it causes flight time skew between the DQS and CLK. Therefore, it needs to allow the controller to compensate for the skew of the DQS signal delays to those signals at the LPDDR4 Memory side through its write leveling capability. The write leveling scheme of this IP core follows all the steps stipulated in the JEDEC specification. For more details on write leveling, refer to the JEDEC specification JESD209-4.

2.2.4. Read Preamble Training

For every read operation, the LPDDR4 I/O primitives of the CPNX device must be initialized at the appropriate time to identify the incoming DQS preamble. Upon proper detection of the preamble, the primitive DQSBUF_IVREF extracts a clean DQS signal out of the incoming DQS signal from the memory and generates the DATAVALID output signal that indicates the correct timing window of the valid read data.

The memory PHY generates an internal pulse signal, READ[3:0], to the primitive DQSBUF_IVREF that is used for the above-mentioned operation. In addition to the READ[3:0] input, another input signal READCLKSEL[2:0] and an output signal, BURSTDET, of the DQSBUF_IVREF block are provided to the PHY to accomplish the READ signal positioning.

Due to the DQS round trip delay that includes PCB routing and I/O pad delays, proper positioning of the READ signal with respect to the incoming preamble is crucial for successful read operations. The CPNX DQSBUF_IVREF block supports a dynamic READ signal positioning function called read preamble training that enables the memory PHY to position the READ signal within an appropriate timing window by progressively shifting the READ signal and monitoring the positioning result.

This read preamble training is performed as part of the memory initialization process after the write leveling operation is complete. During the read preamble training, the memory PHY generates the READ[3:0] pulse, positions this signal using READCLKSEL[2:0] and monitors the BURSTDET output of DQSBUF_IVREF for the result of the current position. The READ signal is set high before the read preamble starts. When the READ pulse is properly positioned, the preamble is detected correctly and the BURSTDET goes high. This guarantees that the generated DATAVALID signal indicates the correct read valid time window.

The READ signal is generated in the system clock (SCLK) domain and stays asserted for the total burst length of the read operation.

A minimum burst length of eight on the memory bus is used in the read preamble training process. The memory PHY can determine the proper position alignment when there is not a single failure on BURSTDET assertions during the multiple trials. If there is any failure, the memory PHY shifts the READ signal position and tries again until it detects no BURSTDET failure.

The memory PHY stores the delay value of the successful position of the READ signal for each DQS group. It uses these delay values during a normal read operation to correctly detect the preamble first, followed by the generation of DATAVALID signal.

2.2.5. Read Training

LPDDR4 devices feature a RD DQ Calibration training function that outputs a 16-bit user-defined pattern on the DQ pins. With this feature Memory Controller can perform read training without write operation, so read training always prior to write training. The purpose of read training is to center the internal Vref (DQ) in the read DQ data eye and at the same time allows for timing adjustments of the DQ and DQS signals to meet setup/hold requirements.

2.2.6. Write Training

The LPDDR4-SDRAM uses an un-matched DQS-DQ path to enable high speed performance and save power in the DRAM. As a result, the DQS strobe must be trained to arrive at the DQ latch center-aligned with the Data eye. The SDRAM DQ receiver is located at the DQ pad and has a shorter internal delay in the SDRAM than the DQS signal. The SDRAM DQ receiver will latch the data present on the DQ bus when DQS reaches the latch, and training is accomplished by delaying the DQ signals relative to DQS such that the Data eye arrives at the receiver latch centered on the DQS transition. Memory Vref (DQ) also be trained during write training.

2.2.7. Error Code and debug

After phy initialization finished, the phy_init_success_o will assert with phy_init_error_code_o == 0x00; or phy_init_fail_o will assert with phy_init_error_code_o != 0x00. The [Table 2.1](#) described the error code meaning and debug suggestions.

Table 2.1 Error Code Description

Error Code	Description
0x01	User parameters violation
0x02	Initialization time out Suggestion: Need to Check input clock frequency and Reset signal.
0x03	Vref (CA) training failed Suggestion: 1) Try to run lower CK frequency 2) Check IO assignment 3) Tuning ODT and Drive strength according SIPI simulation 4) Check PCB Layout even faulty soldering
0x04	CBT move out of range Suggest: 1) check if there is a message for bit error 2) Consider using a smaller value of CBT_x in the GUI
0x05	CBT training success but failed when double check Suggestion: Try to use finer granularity of CBT training and check if the information of hardware problem is printed by UART
0x06	Write leveling first sampled point is high, CK and DQS skew maybe have issue. Suggestion: Try to add DDR CK Delay in the GUI
0x07	Write leveling cannot find rising edge, CK and DQS skew maybe have issue. Suggestion: 1) Try to reduce DDR CK Delay in the GUI and check if the information of hardware problem is printed by UART 2) Change value of DQ ODT
0x08	Can not find common Read Preamble between two ranks. Suggestion: Check CS/CKE skew between two ranks
0x09	Read Preamble does not find. Suggestion: 1) Perform write/read operation before initialization success will cause this error 2) Try to use other Read Latency.
0x0A	Read Preamble is found but sometimes lost during test. Suggestion: 1) Evaluate the PLL Jitter, try to use high PLL VCO parameter 2) Use dictated PLL input I/O, avoid noise on PCB 3) Use lower CK frequency
0x0B	Read training failed due to not enough slack. Suggestion: 1) Try to run lower CK frequency 2) Tuning ODT and Drive strength according SIPI simulation
0x0C	Read training success but failed when double check. Suggestion: 1) Try to use finer granularity of Read training 2) Similar with 0x0B
0x0D	Write training move out of range Suggestion: Try to run lower frequency
0x0E	Reserved
0x0F	Write training failed due to not enough slack. Suggestion: 1) Try to run lower CK frequency 2) Tuning ODT and Drive strength according SIPI simulation
0x10	Vref (DQ) training failed Suggestion: 1) Try to run lower CK frequency 2) Tuning ODT and Drive strength according SIPI simulation
0x11	Internal Error

2.3. LPDDR4 Controller Module

The Controller is considered the high-level management and scheduling to the PHY through DFI, as meanwhile it also included user interface. The controller manages refresh time, address mapping, and schedules BANK/ROW active and precharge, burst writing and reading etc.

2.3.1. Initialization

LPDDR4 memory devices must be initialized before the memory controller can access them. The memory controller starts the memory initialization sequence when the `phy_init_start_i` signal is asserted by the user interface. The `phy_init_start_i` signal needs to be held high. The output signal `phy_init_success_o` or `phy_init_fail_o` is asserted High indicating that the core has ended the initialization sequence, if as Figure 2.2 shows `phy_init_success_o` assert it indicate that ready to access the memory. If as Figure 2.3 shows `phy_init_fail_o` assert, user needs to read `phy_init_error_code_o` to analyze the reason, detail please refers to [Error Code and debug](#).

Before `phy_init_start_i` asserted, user can configures CSR of IP through APB BUS, detail please refer to [Registers Description](#).

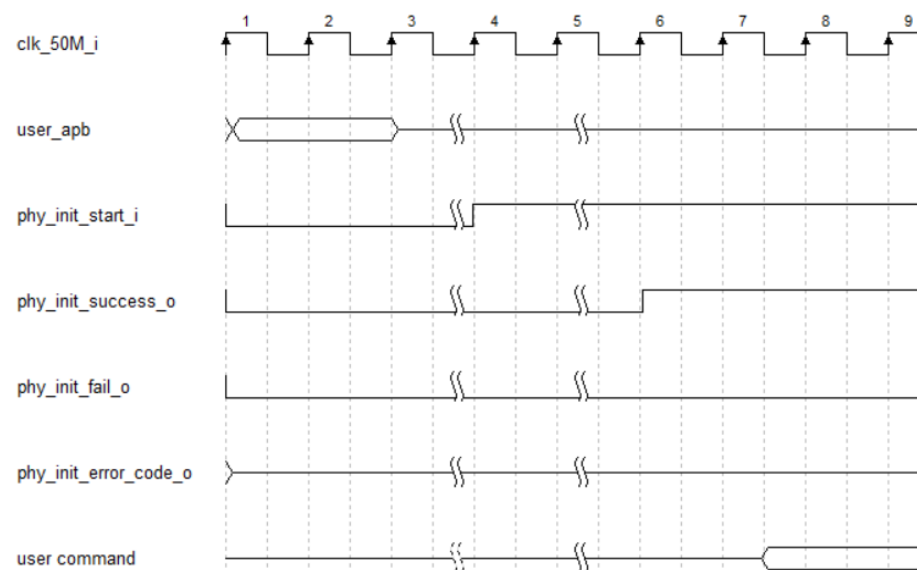


Figure 2.2 Initialization success

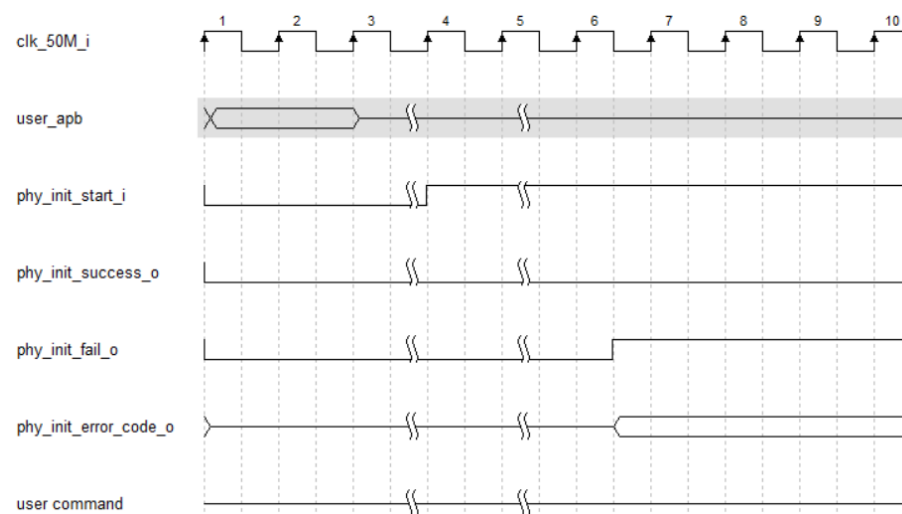


Figure 2.3 Initialization fail

2.3.2. Refresh and Self-Refresh Management

LPDDR4 memory controller core supports eight Auto Refresh commands in one burst. The core has an internal auto refresh generator that sends out a set of consecutive Auto Refresh commands to the memory at once when it reaches the period of the refresh intervals (tREFI) times the Auto refresh burst count selected in user interface.

- Support all bank refresh, not support per bank refresh.
- Only support 8 rows continued refresh.
- Support fixed Refresh rate

Self-Refresh and power down mode allow the system to run under extreme low power meanwhile avoid memory data lost caused by leakage of electricity. Figure 2.4 shows how to enter self-refresh and power down mode as well as how to exit self-refresh and power on.

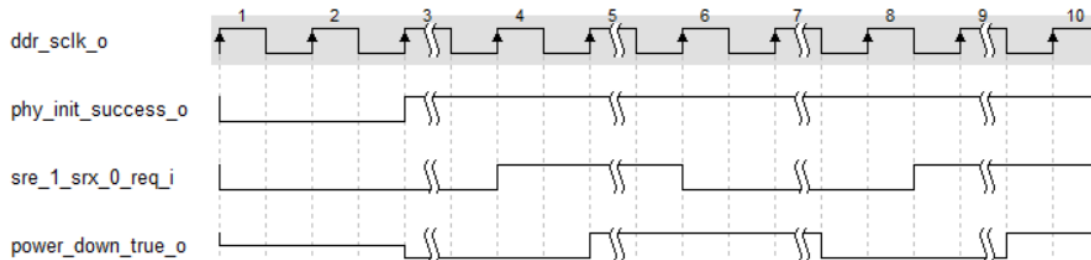


Figure 2.4 Self-Refresh Enter and Exit

2.3.3. Local-to-Memory Address Mapping

Mapping local addresses to memory addresses is an important part of a system design when a memory controller function is implemented. Different with AXI/Avalon BUS, the LPDDR4 Memory Controller address(cmd_addr_i) is addressed in units of the DQ BUS width, each increment points to the next memory location whose size matches the LPDDR4's data bus width. User must know how the local address lines from the memory controller connect to those address lines from the memory because proper local-to-memory address mapping is crucial to meet the system requirements in applications such as a video frame buffer controller. Even for other applications, careful address mapping is generally necessary to optimize the system performance. Figure 2.5 shows the local-to-memory address mapping of the Lattice LPDDR4 memory controller cores.

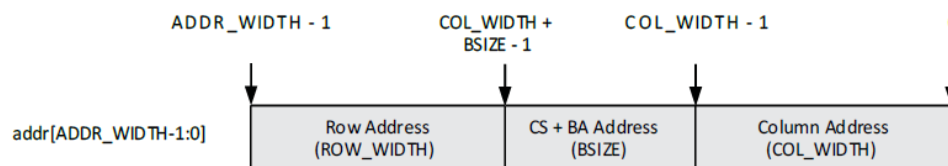


Figure 2.5 Local-to-Memory Address Mapping for Memory Access

ADDR_WIDTH is calculated by the sum of COL_WIDTH, ROW_WIDTH and BSIZE. BSIZE is determined by the sum of the BANK_WIDTH and CS_WIDTH. The value of CS_WIDTH is 1 for a dual-rank configuration and 0 for a single-rank configuration. For LPDDR4 devices, the bank address size is always 3, the column address size is always 10, the single rank system the BSIZE is 3, dual rank system the BSIZE is 4.

2.3.4. Burst and Address Management

When the core is in the command burst operation, it extensively occupies the data bus. During this time, the core prevents cmd_rdy from being asserted until the command burst is completed. While the core is operating in the command burst mode, it can keep maximum throughput by internally replicating the command. The memory controller repeats the given READ or WRITE command up to 64 times. The cmd_burst_count_i [5:0] input is used to set the number of repeats of the given command. The core allows the command burst function to access the memory addresses within the current row. When the core reaches the boundary of the current row while accessing the memory

in the command burst mode, the next address that the core accesses become the beginning of the same row. It causes overwriting the contents of the location or reading unexpected data. Therefore, user must track the accessible address range in the current row while the command burst operation is performed. If an application requires a fixed command burst size, use of 2, 4, 8, 16, 32 or 64 burst count are recommended to ensure that the command burst accesses do not cross the page boundary.

Below are some Address rules.

- Write start address aligned with 16n (C0-C3 always 4'b0000) for BL16.
- Read start address aligned with 4n (C0-C1 always 2'b00)
- When column address over range in one read BL16, remain length will wrap from column 0, next read BL16 will follow the last address. For example, read address is 10'd1020(10'b11_1111_1100), the BL16 will over range, and next BL16 will start from 12 (1020-1024+16)

Burst Management

- Support write and read burst count from 1-64. because $64 * BL16 = 1024$
- Write Mask command only support burst count = 1
- Support burst column address skip. When burst_step = 2'b00, column address increment as 0,16,32,48,...; When burst_step = 2'b01, column address increment as 0,32,64,96,...; When burst_step = 2'10 column address increment as 0,48,96,144,...; when burst_step = 2'11 column address increment as 0,64,128,192,...; This feature is mainly for random access which applies to video rotation.

2.3.5. Local Bus Timing

Local Bus is a low latency memory controller interface, Figure 2.6 shows the command ports, the user command only be accepted when both 'cmd_valid_i' and 'cmd_rdy_o' are assert at same time, as Figure 2.6 shows the user commands only be accepted in 4th sclk and 8th sclk.

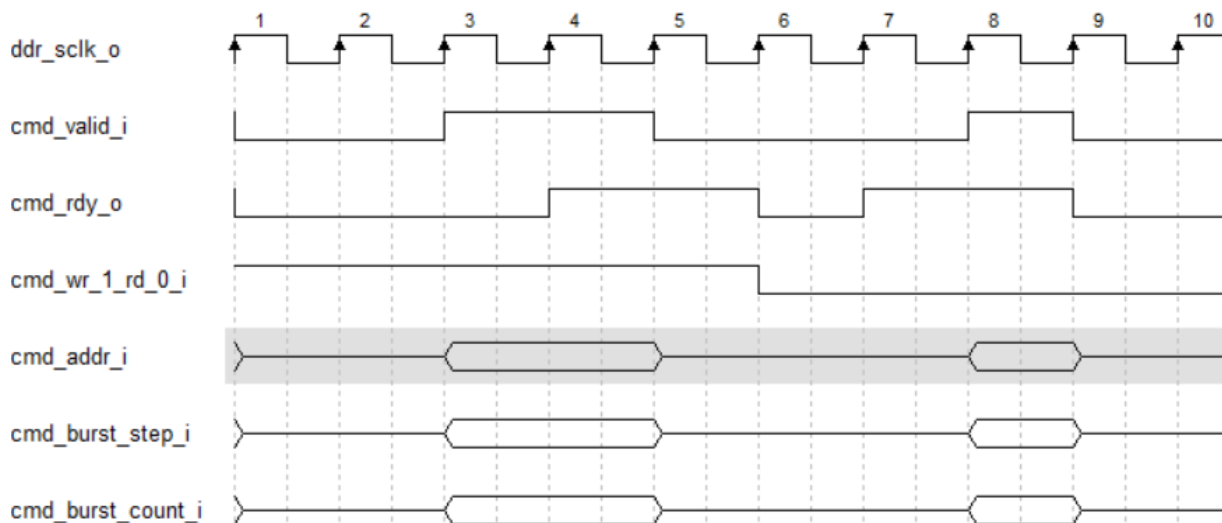


Figure 2.6 Local Bus Command Port

After user write command be accepted, IP will assert datain_rdy_o signal indicate user needs to input write data, after user read command be accepted, IP will assert read_data_valid_o signal means during this period the rd_data_o is valid. Figure 2.7 shows the wr_data_i and datain_rdy_o relationship, rd_data_o and read_data_valid_o relationship.

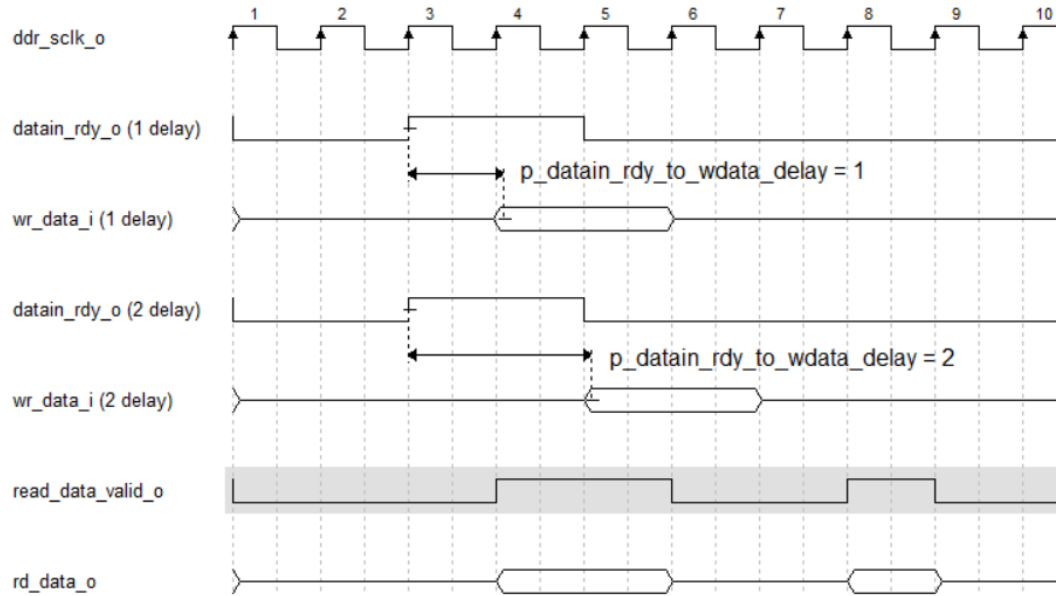


Figure 2.7 Local Bus Data Port

2.3.6. Write Operation

User initiate a memory write operation by asserting `cmd_valid_i` along with the WRITE command and the address. After the WRITE command is accepted, the memory controller core asserts the `datain_rdy_o` signal when it is ready to receive the write data from the user logic to write into the memory. Since the duration from the time a write command is accepted to the time the `datain_rdy_o` signal is asserted is not fixed, the user logic needs to monitor the `datain_rdy_o` signal. Once `datain_rdy_o` is asserted, the core expects valid data on the `wr_data_i` bus one or two clock cycles after the `datain_rdy_o` signal is asserted. User can program the write data delay by setting desired value for `p_datain_rdy_to_wdata_delay` in the user interface, providing flexible backend application support. For example, setting the value to 2 ensures that the core takes the write data in proper time when the local user interface of the core is connected to a synchronous FIFO module inside the user logic. Figure 2.8 shows two examples of the local user interface data write timing.

The controller decodes the `addr` input to extract the current row and current bank addresses and checks if the current row in the memory device is already opened. If there is no opened row in current bank, an ACTIVE command is generated by the controller to the memory to open the current row first. Then the memory controller issues a WRITE command to the memory. If there is already an opened row in the current bank and the current row address is different from the opened row, a PRECHARGE command is generated by the controller to close opened row in the bank. This is followed with an ACTIVE command to open the current row. Then the memory controller issues a WRITE command to the memory. If current row is already opened, only a WRITE command (without any ACTIVE or PRECHARGE commands) is sent to the memory.

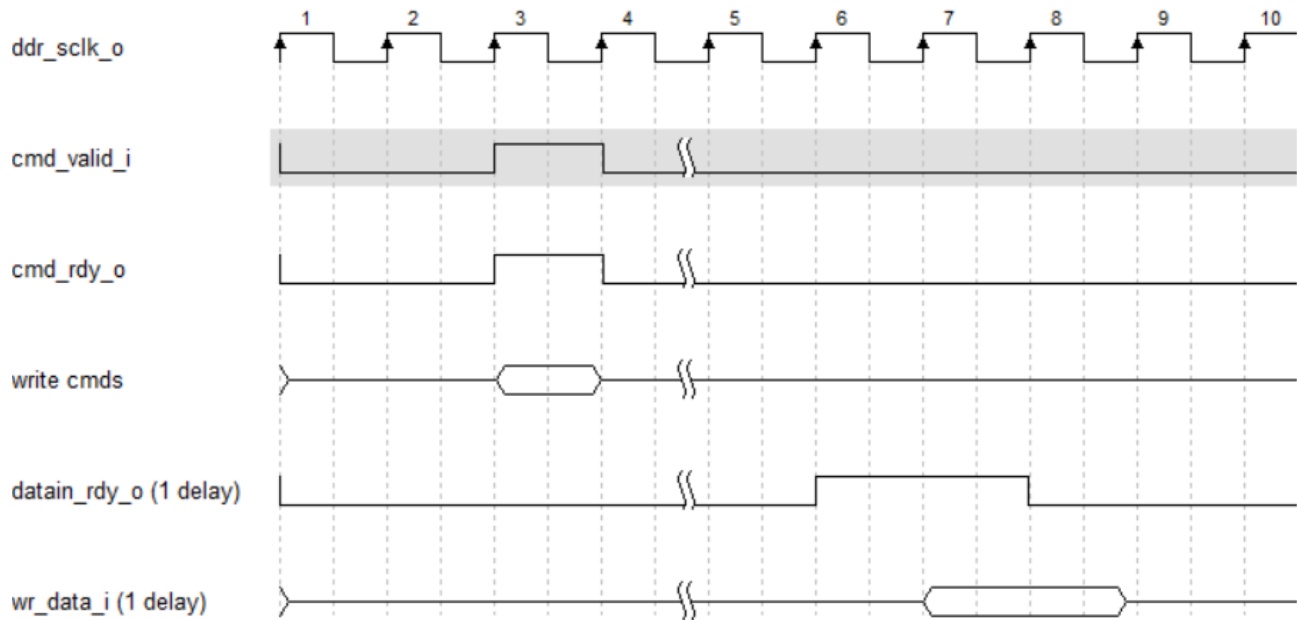


Figure 2.8 Write Operation

2.3.7. Mask Write Operation

The write data mask function to mask the byte lanes for byte-level write. When user enable IP mask write function, the 'cmd_wr_1_rd_0_i' port will extend to 2 bits and there will be three kinds of command read command, write command and mask write command, detail please refer Table 4.1. The mask write command followed timing tCCDMW which longer than tCCD will lower operation efficiency.

When enable mask write function the port 'wr_data_mask_i' port is available, and each bit masks a corresponding byte of local 'wr_data_i'. Under mask write command the 'burst_count_i' port will be ignored by IP, no matter what value inputted by user, IP will use burst count qual 1. As Figure 2.9 shown is mask write timing diagram.

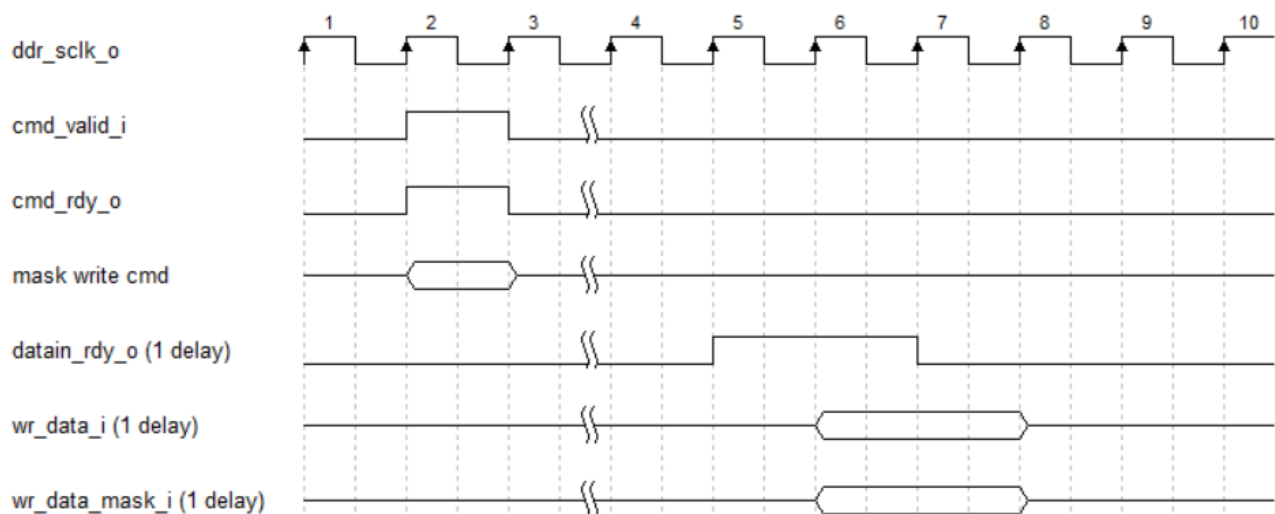


Figure 2.9 Mask Write Operation

2.3.8. Read Operation

When the READ command is accepted, the memory controller core accesses the memory to read the addressed data and brings the data back to the local user interface. Once the read data is available on the local user interface, the memory controller core asserts the `read_data_valid_o` signal to tell the user logic that the valid read data is on the

rd_data_o bus. The read data timing on the local user interface is shown in [Figure 2.10](#).

Read operation follows the same row status checking scheme as mentioned in write operation. Depending on current row status the memory controller generates ACTIVE and PRECHARGE commands as required. More details please refer to the description mentioned in [Write Operation](#).

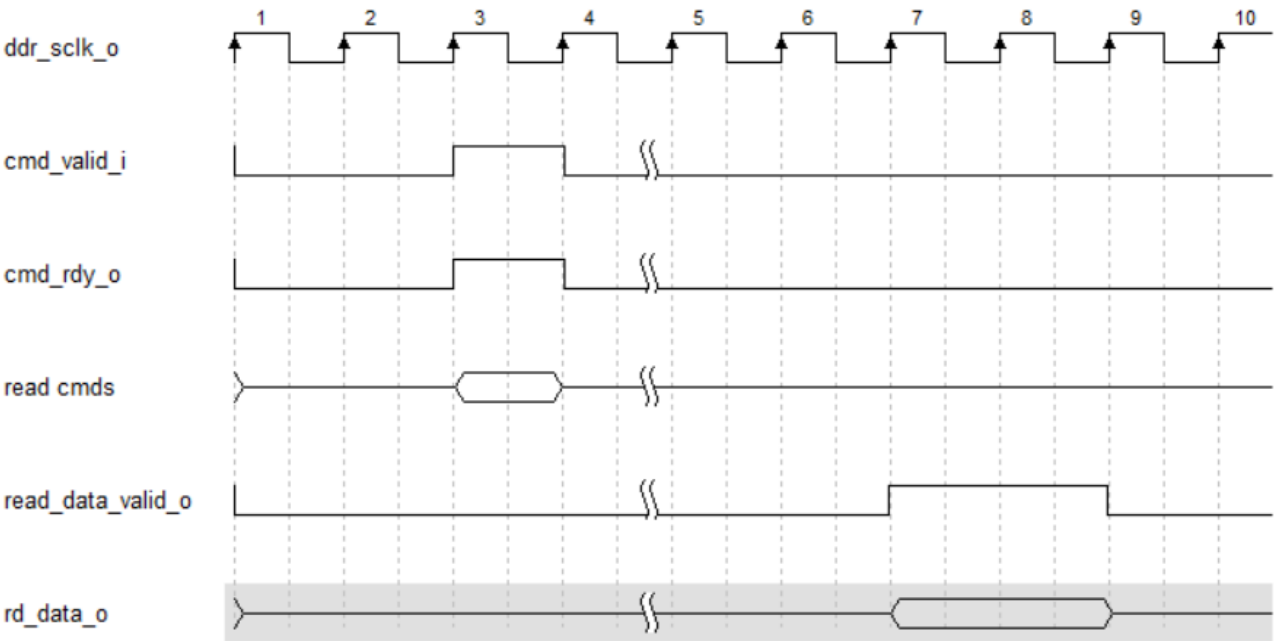


Figure 2.10 Read Operation

3. IP Parameter Description

Table 3.1 Controller Attributes Setting, Table 3.2 Memory Attributes Setting, Table 3.3 PHY Attributes Setting provides the list of user configurable parameters for the CPNX LPDDR4 Controller IP core. These attributes can be configured through the IP Catalog's Module/IP Block Wizard of the Lattice Radiant software. The numerous LPDDR4 SDRAM Controller parameter options are partitioned across multiple user interface tabs as shown in this chapter.

Table 3.1 Controller Attributes Setting

Attribute	Selectable Value	Default	Description
General			
Device	LFCPNX	LFCPNX	Support CertusPro-NX only
Row Width (Memory Density)	13(Dual Channel-2Gb), 14 (Dual Channel-4Gb), 15 (Dual Channel-8Gb), 16 (Dual Channel-16Gb), 17(Dual Channel-32Gb)	13	Dual Channel 32Gbit: Row width is 17 Dual Channel 24Gbit: Row width is 17 Dual Channel 16Gbit: Row width is 16 Dual Channel 12Gbit: Row width is 16 Dual Channel 8Gbit: Row width is 15 Dual Channel 6Gbit: Row width is 15 Dual Channel 4Gbit: Row width is 14 Dual Channel 2Gbit: Row width is 13 Note that the row width should be +1 for same density when single channel device is used
Rank Width	1,2	1	Rank number, 1 is single rank, 2 is dual rank
Memory Data Bus Size	16,24,32,48,64	32	16bit and 24bit will occupy 1 FPGA HP bank 32bit and 48bit will occupy 2 FPGA HP bank 64bit will occupy 3 FPGA HP bank
ODT Width	1,2,3,4	1	CA ODT pad control
Memory Clock (MHz)	266,400,500,533 ,600,700,750,800	533	600MHz, 700MHz, 750MHz and 800MHz only used for stress test, IP does not guarantee the reliability
RefClock (MHz)	20,25,50,100	100	PLL Reference Clock frequency, suggest using high frequency clock to reduce PLL Jitter; 20MHz reference clock can't support 750MHz Memory clock
burst_step_00 definition(column)	16	16	When cmd_burst_step_i is 2'b00, during burst write/read the column address will be increased 16 automatically, this setting is fixed;
burst_step_01 definition(column)	32	32	When cmd_burst_step_i is 2'b01, during burst write/read the column address will be increased by user_define_value (This value must be a multiple of 16). For example, if user_define_value is 32, and burst begin address is 16, the data will be write/read from/to address 16+BL16, then 48+BL16, then 80+BL16.....
burst_step_10 definition(column)	48	48	When cmd_burst_step_i is 2'b10, during burst write/read the column address will be increased by user_define_value (This value must be a multiple of 16). For example, if user_define_value is 48, and burst begin address is 16, the data will be write/read from/to address 16+BL16, then 64+BL16, then 112+BL16.....
burst_step_11 definition(column)	64	64	When cmd_burst_step_i is 2'b11, during burst write/read the column address will be increased by

			user_define_value (This value must be a multiple of 16). For example, if user_define_value is 64, and burst begin address is 16, the data will be write/read from/to address 16+BL16, then 80+BL16, then 144+BL16.....
Interface			
Enable APB Interface	Enable, Disable	Enable	User access CSR through APB Bus to realize debugging purpose, this interface is optional
Enable UART Interface	Enable, Disable	Enable	UART TXD prints the initialization result and calibration detail like Vref margin and timing margin during training steps mentioned in LPDDR4 PHY Module ; UART RXD function is reserved.
Enable DMI Interface	Enable, Disable	Enable	If disable DMI IO, need pull-down LPDDR4 SDRAM DMI IO on PCB.
Enable ODT Interface	Enable, Disable	Enable	If disable ODT IO, need pull-up or pull-down LPDDR4 SDRAM CA ODT IO
Enable Self-Refresh Interface	Enable, Disable	Enable	Enable the self-refresh and power down function
Enable Interrupt Interface	Enable, Disable	Enable	Enable interrupt interface, more detail please refer to Interrupt Register
Timing			
datain_rdy_o to wr_data_i delay(sclk)	1,2	2	When IP core assert datain_rdy_o need wait 1 or 2 sclk to input write data
tRFCab(sclk)	18	Calculated	Refresh Cycle Time: Delay between the REFRESH command and the next valid command, except DES command
tREFlab(sclk)	521	Calculated	Average Refresh Interval (All bank)
tRPpb(sclk)	3	Calculated	Row precharge time (single bank)
tRPab(sclk)	3	Calculated	Row precharge time (all banks)
tRTP(sclk)	2	Calculated	Internal READ to PRECHARGE command delay
tWR(sclk)	3	Calculated	WRITE recovery time: After a burst WRITE operation, tWR must be satisfied before a PRECHARGE command to the same bank can be issued.
tRRD(sclk)	2	Calculated	Active bank-A to active bank-B: The minimum time interval between ACTIVATE commands to different banks
tRCD(sclk)	3	Calculated	RAS-to-CAS delay: The device can accept a READ or WRITE command at tRCD after the ACTIVATE command is issued
tRAS(sclk)	6	Calculated	Row active time: The minimum time interval from ACTIVATE command to Precharge
tCCD(sclk)	3	Calculated	CAS-to-CAS delay
tCCDMW(sclk)	8	Calculated	A Masked Write command to the same bank cannot be issued until tCCDMW later, to allow the LPDDR4-SDRAM to finish the internal Read-Modify-Write
tWTR(sclk)	2	Calculated	write to read delay, dqs post-amble end to read command.

t_READ_to_WRITE(sclk)	5	Calculated	read to write delay
t_XP(sclk)	2	Calculated	Exit Power-Down to next valid command delay
t_XSR(sclk)	39	Calculated	Self-Refresh exit to next valid command delay

Table 3.2 Memory Attributes Setting

Attribute	Selectable Value	Default	Description
MR03_Setting			
PU_CAL	VDDQ/2.5, VDDQ/3	VDDQ/3	Pull-up Calibration Point
PDDS	RZQ/1, RZQ/2, RZQ/3, RZQ/4, RZQ/5, RZQ/6,	RZQ/6	Pull-Down Drive Strength
DBI-RD	"Disable"	"Disable"	DBI-Read Function Disable; Display only
DBI-WR	"Disable"	"Disable"	DBI-Write Function Disable; Display only
MR11_Setting			
CA ODT	RZQ/1, RZQ/2, RZQ/3, RZQ/4, RZQ/5, RZQ/6, Disable	RZQ/1	CA Bus Receiver On-Die-Termination
DQ ODT	RZQ/1, RZQ/2, RZQ/3, RZQ/4, RZQ/5, RZQ/6, Disable	RZQ/6	DQ Bus Receiver On-Die-Termination
MR13_Setting			
Data Mask Enable	"Enable", "Disable"	"Disable"	Mode Register Data Mask setting, detail please refer Table 6.1
MR22_Setting			
CA-ODT Enable	"Enable", "Disable"	"Enable"	CA ODT termination enable
CS-ODT Enable	"Enable", "Disable"	"Enable"	CS ODT enable for nonterminating rank
CK-ODT Enable	"Enable", "Disable"	"Disable"	CK ODT enabled for nonterminating rank
FPGA ODT	RZQ/1, RZQ/2, RZQ/3, RZQ/4, RZQ/5, RZQ/6, Disable	RZQ/6	Controller ODT Value for VOH calibration

Table 3.3 PHY Attributes Setting

Attribute	Selectable Value	Default	Description
General			
Simulation Mode Enable	"Enable", "Disable"	"Disable"	Enable simulation mode to shorten the initialization and training procedure
Vref (CA/DQ) Training Enable	"Enable", "Disable"	"Enable"	Including Vref (CA)training, FPGA Vref(DQ) training, Memory Vref(DQ) training
Initialization Enable	"Enable", "Disable"	"Enable"	Including tINIT1~tINIT5 defined in JESD209-4 and ZQ calibration
Command Bus Training Enable	"Enable", "Disable"	"Enable"	Including CK CS CA calibration, and Vref (CA) training
Write Leveling Enable	"Enable", "Disable"	"Enable"	Per DQS Group write leveling training
Read Preamble Training Enable	"Enable", "Disable"	"Enable"	Read Preamble detect training
Read Training Enable	"Enable", "Disable"	"Enable"	Including read DQ delay calibration and FPGA Vref (DQ) training
Write Training Enable	"Enable", "Disable"	"Enable"	Including write DQ delay calibration and Memory Vref (DQ) training

Advanced			
DDR CLK Delay (Steps)	[0-128]	75	Used for compensating CK and DQS skew, maybe need tuning according to PCB design.
CBT Move Step (X)	1,2,4,8	4	CA Delay calibration granularity, finer granularity cost more time but fine training result.
CBT Move Step (Y)	1,2,4,8	4	Vref(CA) training granularity, finer granularity cost more time but fine training result.
Read Training Move Step (X)	1,2,4,8	4	Read DQ Delay calibration granularity, finer granularity cost more time but fine training result.
Read Training Move Step (Y)	1,2,4,8	4	Read DQ training granularity, finer granularity cost more time but fine training result.
Write Training Move Step (X)	1,2,4,8	4	Write DQ Delay calibration granularity, finer granularity cost more time but fine training result.
Write Training Move Step (Y)	1,2,4,8	4	Write DQ training granularity, finer granularity cost more time but fine training result.

4. Signal Descriptions

Table 4.1 Port Definitions shows the port definitions of the CertusPro-NX LPDDR4 Memory Controller IP.

Table 4.1 Port Definitions

Port name	I/O	Width	Description
Clock and Reset			
clk_50M_i	input	1	This Clock used for RISC-V, APB Bus, GDDR SYNC, suggest 50MHz
ddr_ref_clk_i	input	1	PLL Reference Clock input
rst_n_i	input	1	Asynchronous active low reset
ddr_sclk_o	output	1	System clock. This is ¼ of the DDR clock frequency and is the main clock of the LPDDR4 Memory Controller IP.
pll_lock_o	output	1	PLL lock output indicating when PLL is locked
Initialization (clk_50M_i domain)			
phy_init_start_i	input	1	After power on the training engine monitors this signal, once detect this signal asserted, will read User APB configuration then start memory initialization and calibration; It is suggested to keep this signal on a high level
phy_init_success_o	output	1	Initialization and Calibration success, user can write read memory
phy_init_fail_o	output	1	Initialization and Calibration fail, user can't write read memory.
phy_init_error_code_o	output	1	When initialization is successful the error code is 0x00. When initialization fail, please find the error code definition in Table 2.1
UART (clk_50M_i domain)			
ip_uart_en_o	output	1	When assert means training engine is printing information. Sometimes user UART and IP UART share same IO, and use this signal as multiplexing
uart_rxd_i	input	1	UART Receiver
uart_txd_o	output	1	UART Transmitter
APB BUS (clk_50M_i domain)			
user_apb_psel_i	input	1	APB Select signal
user_apb_penable_i	input	1	APB Enable signal
user_apb_pwrite_i	input	1	APB Direction signal; 1 = Write, 0 = Read
user_apb_paddr_i	input	10	APB Address
user_apb_pwdata_i	input	32	APB Write data
user_apb_prdata_o	output	32	APB Read data
user_apb_pready_o	output	1	APB Ready
mc_interrupt_o	output	1	Memory Controller Interrupt, read the interrupt register will clear interrupt signal. The detail please refer Interrupt Register
Control I/F (ddr_sclk_o domain)			
cmd_valid_i	input	1	Command and address valid input. When asserted, the addr, cmd, ofly_burst_len and cmd_burst_cnt inputs are considered valid.
cmd_rdy_o	output	1	Command ready output – When asserted, indicates that the core is ready to accept the next command and the corresponding address.
cmd_wr_1_rd_0_i	input	1 or 2	User command input to the memory controller.

			When disable write mask function this port is 1 bit 1'b0 is read command 1'b1 is write command when enable write mask function this port is 2bit 2'b00 is read command 2'b01 is write command 2'b10 is reserved for future use 2'b11 is mask write command
cmd_addr_i	input	Row_Width + 3 + 10 (Single RANK); Row_Width + 1 + 3 + 10 (Dual RANK)	Read or write address controlled by user to the memory controller. According JESD209-4 the write address should be 16n which means the cmd_addr_i[3:0] should be always 4'b0000; the read address should be 4n which means the cmd_addr_i[1:0] should be always 2'b00. Otherwise, the low bit address will be ignored Refer to Local-to-Memory Address Mapping for more details.
cmd_burst_step_i	input	2	2'b00: When burst count more than 1, column address increment by 16. The address of repeated BL16 write/read commands is continued. Like 0, 16, 32, 48, ...1024 2'b01: When burst count more than 1, column address increment by 32. The address of repeated BL16 write/read commands is spaced. Like 0, 32, 64, 96, ...1024, this means address 16~31 is not written or read. 2'b10: When burst count more than 1, column address increment by 48. The address of repeated BL16 write/read commands is spaced. Like 0, 48, 96, 144, ...1024, this means address 16~48 is not written or read. 2'b11: When burst count more than 1, column address increment by 64. The address of repeated BL16 write/read commands is spaced. Like 0, 64, 128, 192, ...1024, this means address 16~64 is not written or read.
cmd_burst_count_i	input	6	Command burst count input – Indicates the number of times a given read or write command is to be repeated by the controller automatically. 0 means 1* BL16, 1 means 2* BL16, 63 means 64* BL16. Due to MC side is 1:8 but Memory side is BL16, so each BL16 write will assert two sclk cycle datain_rdy_o, each BL16 read will assert two sclk cycle read_data_valid_o. The command burst count only repeats write command and read command, not repeat mask write command. When user input mask write command IP will ignore command burst count input value.
sre_1_srx_0_req_i	input	1	1'b1: Ask LPDDR4 enter or keep self-refresh and power down mode 1'b0: Ask LPDDR4 exit self-refresh and power on or keep power on mode Asking self-refresh the write/read operation will be not allowed, the 'cmd_rdy' signal will not assert until self-refresh exit.
power_down_true_o	output	1	1'b0: Power on state

			1'b1: Power down state
ca_odt_i	input	odt_width	CA ODT pad control, user needs to control this signal according to the hardware setting
Data I/F (ddr_sclk_o domain)			
datain_rdy_o	output	1	Data ready output – When asserted, indicates the core is ready to receive the write data. There is 1 or 2 sclk delay between datain_rdy_o to wr_data_i
wr_data_i	input	dq_width*8	Write data input from user logic to the memory controller. The user side write data width is x8 times the memory data bus.
wr_data_mask_i	input	dq_width	Only available when user enable write mask function. After send write mask command, user needs to input wr_data_mask_i along with wr_data_i.
read_data_valid_o	output	1	Read data valid output – When asserted, indicates the data on the read_data bus is valid.
rd_data_o	output	dq_width*8	Read data output from memory controller to the user logic.
LPDDR4 Memory Interface			
em_ddr_ck_o	out	1	LPDDR4 CK Signal
em_ddr_cke_o	out	rank_width	LPDDR4 CKE Signal
em_ddr_cs_o	out	rank_width	LPDDR4 CS Signal
em_ddr_ca_o	out	6	LPDDR4 CA Signal
em_ddr_reset_n_o	out	1	LPDDR4 Reset Signal
em_ddr_dq_io	inout	dq_width	LPDDR4 DQ Signal
em_ddr_dqs_io	inout	dq_width/8	LPDDR4 DQS Signal
em_ddr_dmi_io	inout	dq_width/8	LPDDR4 DMI Signal
em_ddr_ca_odt_o	out	odt_width	LPDDR4 CA ODT Signal

5. Registers Description

Table 5.1 Register Map

Offset	Register Name	Access Type	Description
0x00	Feature Control Register 0	WO	Memory configuration register
0x04	Feature Control Register 1	WO	Training control register
0x08	Status Register 0	RO	Training configuration register
0x0C	Status Register 1	RO	Clock setting register
0x10	Status Register 2	RO	Training result register
0x88	Interrupt Register	WR	Interrupt register
Others	Internal Register	Reserved	-

Table 5.1 Register Map describes the register map. This chapter will describe the registers, which are available for user to configure memory controller IP. To be mentioned, these registers are already configured by the GUI settings during user' instantiation. However, for advanced debug and development, it's an optional way for user to configure these registers through USER APB BUS. For detailed timing diagram, please refer to [Figure 2.2](#).

5.1. Feature Control Register 0

The Feature Control Register 0 is provided to configure DDR CLK delay and some of LPDDR4 memory mode registers. The definition of each bit is presented in [Table 5.2](#) Feature Control Register 0 Information(user_apb_paddr_i=00H).

Table 5.2 Feature Control Register 0 Information(user_apb_paddr_i=00H)

Field	Name	Access Type	Width	Reset	Description
[31]	Reserved	—	1	—	—
[30:24]	DDR CLK delay	WR	7	75	[0,128]
[23]	Reserved	—	1	—	—
[22]	DBI-RD enable	WR	1	0	0: Disable (not available in Beta version) 1: Enable
[21]	DBI-WR enable	WR	1	0	0: Disable (not available in Beta version) 1: Enable
[20]	Data Mask enable	WR	1	0	0: Disable (not available in Beta version) 1: Enable
[19]	PU-CAL	WR	1	1	0: VDDQ/2.5 1: VDDQ/3
[18:16]	PDDS	WR	3	6	1: RZQ/1 2: RZQ/2 3: RZQ/3 4: RZQ/4

					5: RZQ/5 6: RZQ/6
[15:14]	Reserved	—	2	—	—
[13:11]	FPGA ODT	WR	3	6	0: Disable 1: RZQ/1 2: RZQ/2 3: RZQ/3 4: RZQ/4 5: RZQ/5 6: RZQ/6
[10:8]	DQ ODT	WR	3	6	0: Disable 1: RZQ/1 2: RZQ/2 3: RZQ/3 4: RZQ/4 5: RZQ/5 6: RZQ/6
[7:6]	Reserved	—	2	—	—
[5]	CA ODT enable	WR	1	0	0: enable 1: disable
[4]	CS ODT enable	WR	1	1	0: enable 1: disable
[3]	CK ODT enable	WR	1	1	0: enable 1: disable
[2:0]	CA ODT value	WR	3	1	0: Disable 1: RZQ/1 2: RZQ/2 3: RZQ/3 4: RZQ/4 5: RZQ/5 6: RZQ/6

5.2. Feature Control Register 1

Feature control register 1 reflects the basic settings such as memory clock, data bus size and training steps. The definition of each bit is presented in [Table 5.3](#).

Table 5.3 Feature Control Register 1 Information(user_apb_paddr_i=04H)

Field	Name	Access	Width	Reset	Description
[31:29]	Memory Clock	WR	3	3	The actual frequency is determined by the combination of [31:29] and [15] bits. If [15]=0,

					0: 266MHz 1: 400MHz 2: 500MHz 3: 533MHz 4: 600MHz 5: 750MHz 6: 800MHz if [15]=1: 700MHz
[28]	Reserved	—	1	—	—
[27:26]	Read Training Move Step(X-axis)	WR	2	2	0: 1 step 1: 2 steps 2: 4 steps 3: 8 steps
[25:24]	Read Training Move Step(Y-axis)	WR	2	2	0: 1 step 1: 2 steps 2: 4 steps 3: 8 steps
[23:20]	Memory Data Bus Size	WR	4	4	2: 16 bits 3: 24 bits 4: 32 bits 6: 48 bits 8: 64 bits
[19:18]	Write Training Move Step(X-axis)	WR	2	2	0: 1 step 1: 2 steps 2: 4 steps 3: 8 steps
[17:16]	Write Training Move Step(Y-axis)	WR	2	2	0: 1 step 1: 2 steps 2: 4 steps 3: 8 steps
[15]	Memory Clock Extension	WR	1	0	This bit is the extension of the bit [31:29]. If this bit is set to 1, the Memory Clock in bit [31:29] is 3'b000, the actual frequency is 700MHz (1400Mbps)
[14]	Reserved	—	1	—	—
[13:12]	CBT Move Step(X-axis)	WR	2	2	0: 1 step 1: 2 steps 2: 4 steps 3: 8 steps

[11:10]	CBT Move Step(Y-axis)	WR	2	2	0: 1 step 1: 2 steps 2: 4 steps 3: 8 steps
[9]	Reserved	—	1	—	—
[8]	UART Print Detail of Initialization & Calibration Enable	WR	1	0	0: Disable 1: Enable,
[7]	Simulation Mode Enable	WR	1	0	0: Disable 1: Enable
[6]	Vref(CA/DQ) Training Enable	WR	1	1	0: Disable 1: Enable
[5]	Write Training Enable	WR	1	1	0: Disable 1: Enable
[4]	Read Training Enable	WR	1	1	0: Disable 1: Enable
[3]	Read Preamble Training Enable	WR	1	1	0: Disable 1: Enable
[2]	Write Leveling Enable	WR	1	1	0: Disable 1: Enable
[1]	Command Bus Training Enable	WR	1	1	0: Disable 1: Enable
[0]	Initialization Enable	WR	1	1	0: Disable 1: Enable

5.3. Status Register 0

Status register 0 reflects training parameters such as Vref (DQ), read clock select and read DQS latency. The definition of each bit is presented in [Table 5.4](#).

Table 5.4 Status Register 0 Information(user_apb_paddr_i=08H)

Field	Name	Access	Width	Reset	Description
[31:29]	Reserved	—	3	—	—
[28:22]	DQS_BUF Vref	RO	7	48	Vref (DQ) of FPGA side
[21]	Reserved	—	1	—	—
[20]	Pseudo-Random Bit Sequence Enable	RO	1	1	0: Disable 1: Enable
[19:16]	Read Clock Select	RO	4	13	Parameter of read preamble training
[15]	Reserved	—	1	—	—
[14:5]	Read DQS Latency	RO	10	2	Parameter of read preamble training
[4:2]	Reserved	—	3	—	—

[1]	Command from Training Engine	RO	1	0	0: Command is from controller 1: Command is from training engine
[0]	Write Data from Training Engine	RO	1	0	0: Write data is from controller 1: Write data is from training engine

5.4. Status Register 1

Status Register 1 is a status register of DDR clock, from which the state of DDR clock can be used to diagnose the training. The definition of each bit is presented in [Table 5.5](#).

Table 5.5 Status Register 1 Information(user_apb_paddr_i=0CH)

Field	Name	Access	Width	Reset	Description
[31:5]	Reserve	—	27	—	—
[4]	Clock Update	RO	1	0	A transition from LOW to HIGH triggers the clock frequency change according to the DDR Clock Selected.
[3]	DLL Update	RO	1	0	Enables DLL update during clock frequency change. 0: Disable 1: Enable
[2:0]	DDR Clock Select	RO	3	0	0: Operating frequency 1: Command bus training frequency

5.5. Status Register 2

The status register 2 reflects the training results including training state, error code. The definition of each bit is presented in [Table 5.6](#).

Table 5.6 Status Register 2 Information(user_apb_paddr_i=10H)

Field	Name	Access	Width	Reset	Description
[31]	Training Start	RO	1	0	0: start 1: not start
[30]	Training Error	RO	1	0	0: no error 1: error
[29]	Training Complete	RO	1	0	0: incomplete 1: complete
[28]	Reserve	—	1	—	—

[27:20]	Error Code	RO	8	0	More details please refer to Error Code and debug
[19:12]	Compare Error of Write/Read Training	RO	8	0	Each bit reflects the compare error of its corresponding DQS group 0: no error 1: error
[11]	Reserve	—	1	—	—
[10:2]	DLL Code	RO	9		Generated by DDRDLL, means the move steps of 90-degree phase shift
[1]	Read Burst Detected Signal	RO	1		0: the captured read DQ/DML is unreliable 1: the read preamble and DQS pulses are captured properly
[0]	DDR Clock and SCLK Ready	RO	1		0: not ready 1: ready

5.6. Interrupt Register

The Interrupt Register is comprised of Interrupt enable bits and Interrupt events. In current version there are four interrupt events, and each interrupt event can be enabled or disable individually through programmable interrupt enable bits. Any enabled interrupt event can assert the interrupt signal(mc_interrupt_o) and user reading the interrupt register(0x88) can clear the interrupt signal(mc_interrupt_o). If user needs to utilize the interrupt function, it's suggested to follow this sequence:

1. The user needs to write '1' to enable the correspond bit in the interrupt register via the APB bus.
2. The signal mc_interrupt_o should be used as the interrupt and integrated into the user logic.
3. Once the mc_interrupt_o is asserted, the user should read the interrupt register to identify which interrupt event is asserted in detail. At this point, the interrupt assertion will be cleared upon reading the interrupt register.

[Table 5.7](#) shows the definition of each bit in this register.

Table 5.7 Interrupt Register Information(user_apb_paddr_i=88H)

Field	Name	Access	Width	Reset	Description
[31:20]	Reserve	—	12	—	—
[19]	Interrupt Assertion for Self-refresh and Power Down Exit	RO	1	0	0: interrupt unasserted 1: interrupt asserted
[18]	Interrupt Assertion for Self-refresh and Power Down Entry	RO	1	0	
[17]	Interrupt Assertion for Training fail	RO	1	0	
[16]	Interrupt Assertion for Training Success	RO	1	0	
[15:4]	Reserve	—	12	—	—
[3]	Interrupt Enable for Self-refresh and Power Down Exit	WR	1	0	Write 0: disable the interrupt

[2]	Interrupt Enable for Self-refresh and Power Down Entry	WR	1	0	Write 1: enable the interrupt
[1]	Interrupt Enable for Training Fail	WR	1	0	
[0]	Interrupt Enable for Training Success	WR	1	0	

6. IP Core Generation and Evaluation

This section describes the steps required with Lattice Radiant software to configure and generate the CertusPro-NX LPDDR4 Controller IP. It is recommended to instantiate this IP core in the Lattice Radiant 2023.1 or later. The GUI interface has three tabs as [Figure 6.1](#), [Figure 6.2](#), [Figure 6.3](#).

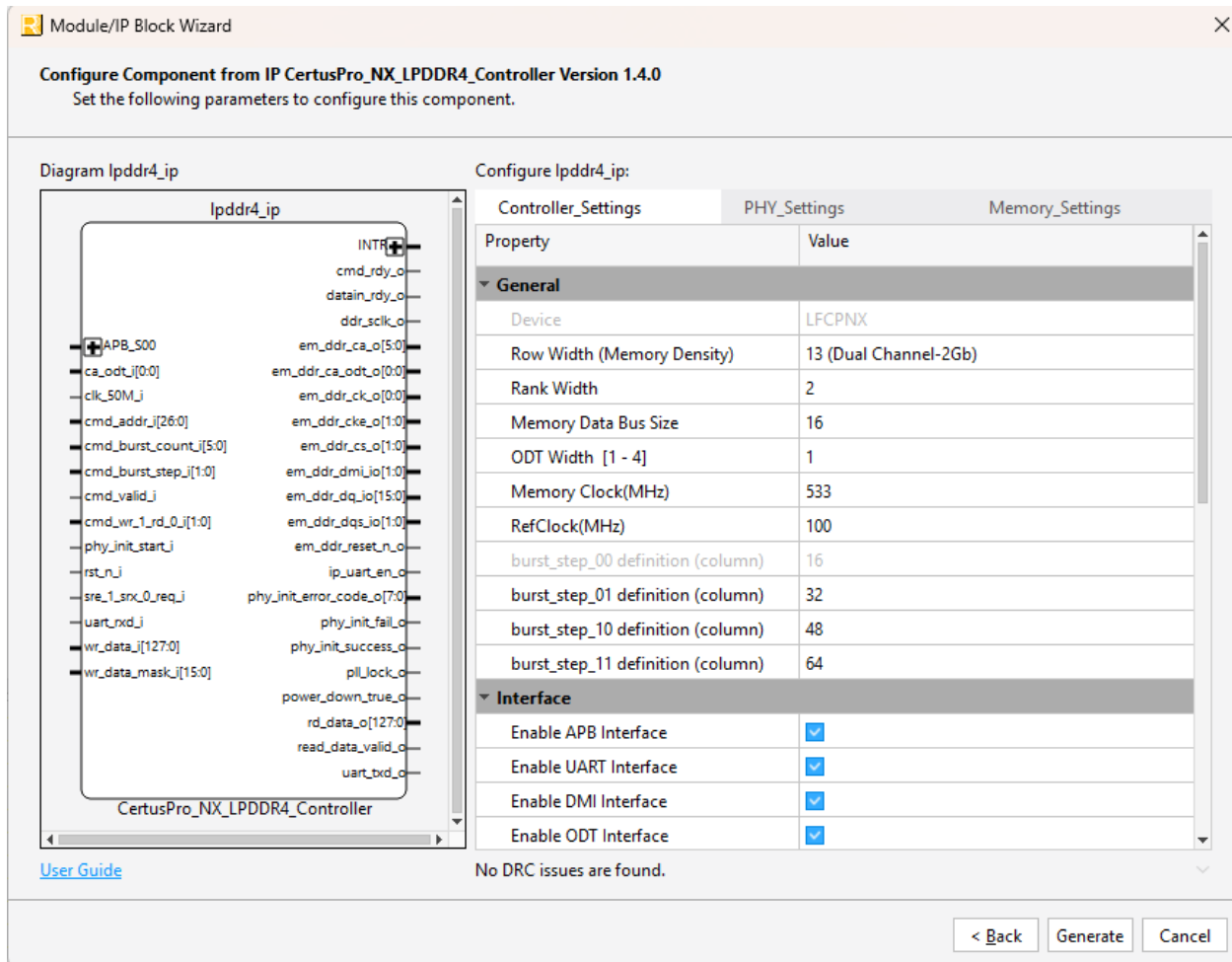


Figure 6.1 GUI Controller Settings

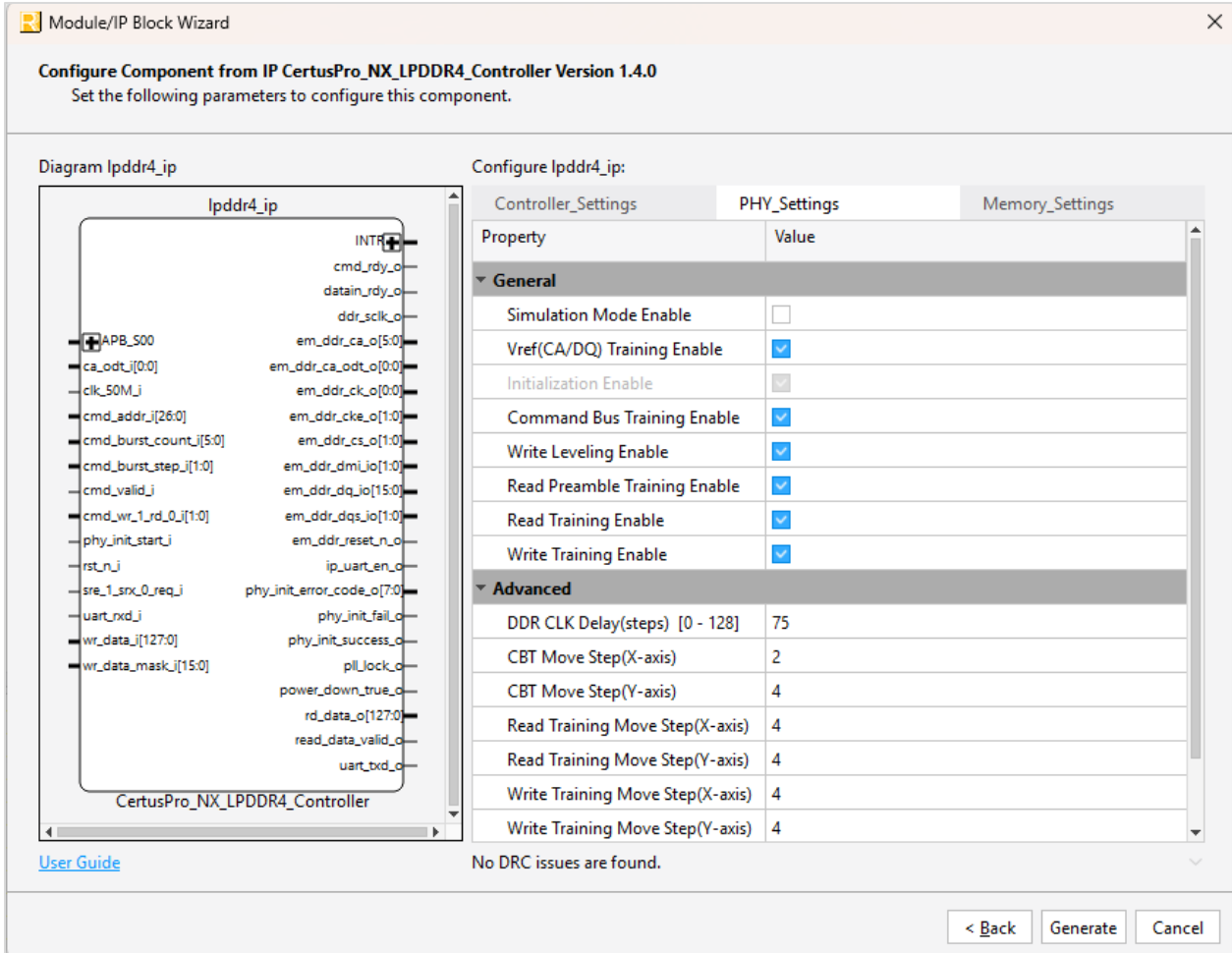


Figure 6.2 GUI PHY Setting

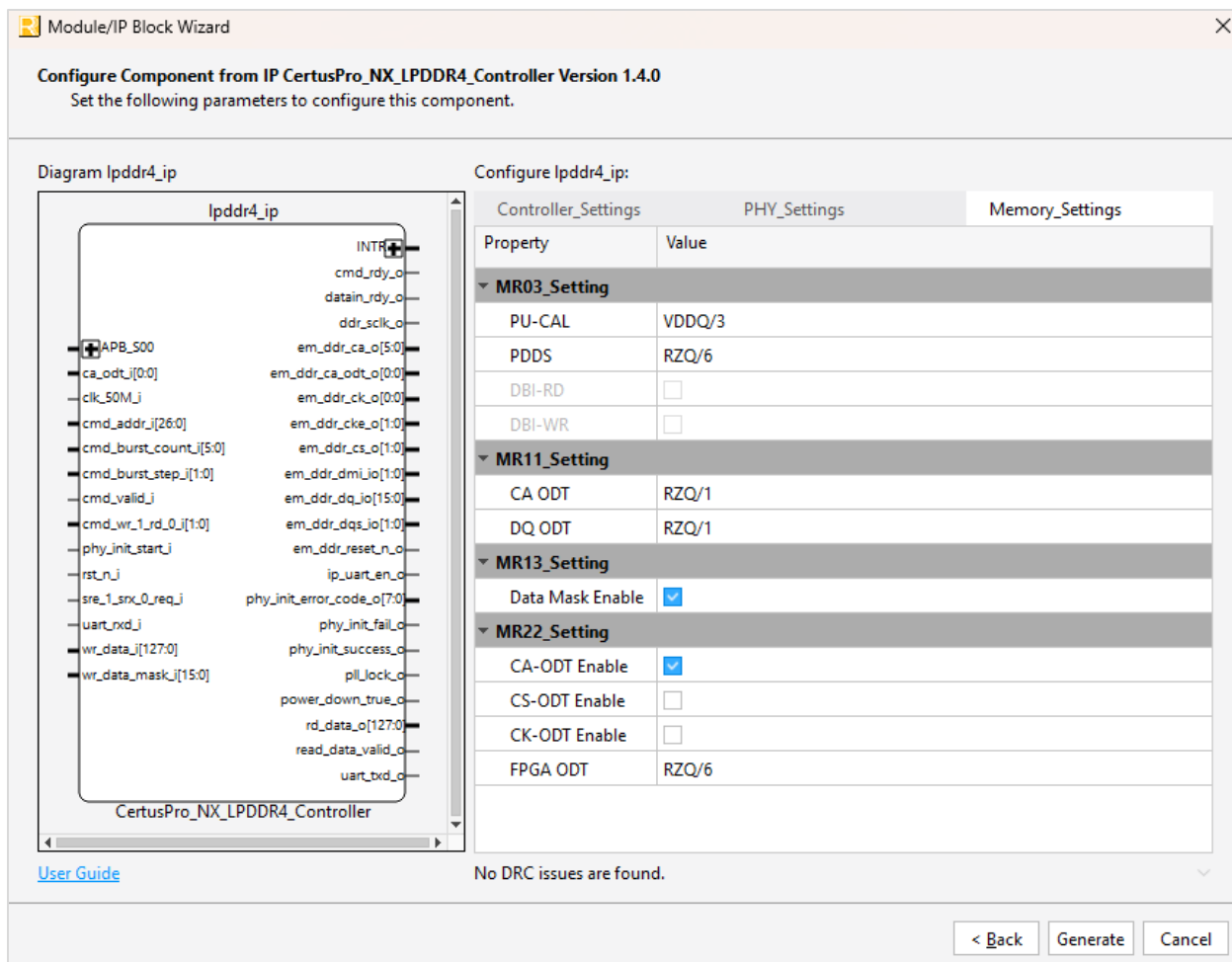


Figure 6.3 GUI Memory Setting

6.1. Mask Write Function Setting

Please note that the user should configure the IP to either enable or disable the mask operation (including mask write/read operations) according to their design. To accommodate these designs, two parameters need to be configured. Figure 6.1 shows setting of 'Enable DMI Interface' and Figure 6.3 shows setting of 'Data Mask Enable', which will enable the DRAM control port and the user-controlled signal of DMI respectively. Table 6.1 is the truth table of IP configuration and related mask operation.

Table 6.1 Truth Table of IP configuration and related mask operation

	Enable DMI Interface checked	Enable DMI Interface unchecked
Data Mask Enable checked	Data Mask function enabled	Illegal setting
Data Mask Enable unchecked	Data Mask function disabled, LPDDR4 Memory DMI pin is pull down by FPGA	Data Mask function disabled, the LPDDR4 Memory DMI pin is pull down by external pull- down resistor. LPDDR4 data mask pin doesn't need drive by FPGA

6.2. Self-Refresh Power Down Function Setting

As Figure 6.4 shows, the function of self-refresh power down is supported. To enable it, the user should check the checkbox in the 'Controller_Settings' tab. Once this function is enabled, the sre_1_srx_0_req_i and power_down_true_o signals will be available. This can be used to control and monitor the state switching, respectively.

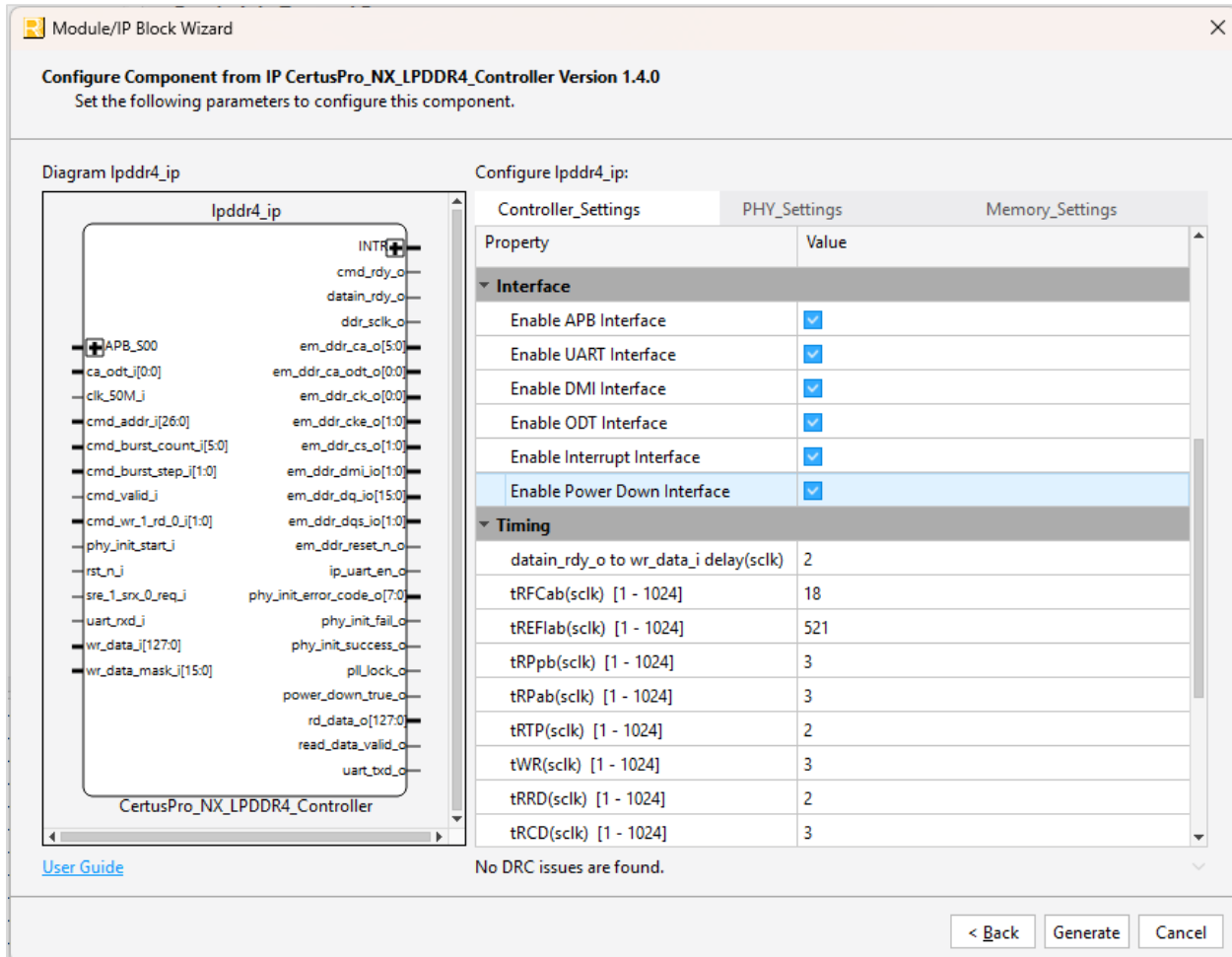


Figure 6.4 Self-Refresh Power Down Function Setting

6.3. Timing Constraint

After generating IP through GUI, the constraint file can be found under the installation directory as Figure 6.5.

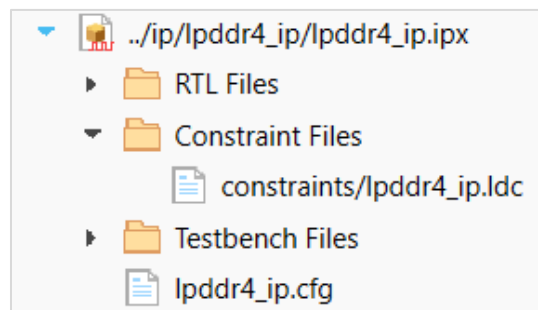


Figure 6.5 Directory of Constraint File

It is recommended to add timing constraint as below steps:

- (1) Finish with synthesize design
- (2) Open Post-Synthesis Timing Constraint Editor as [Figure 6.6](#)

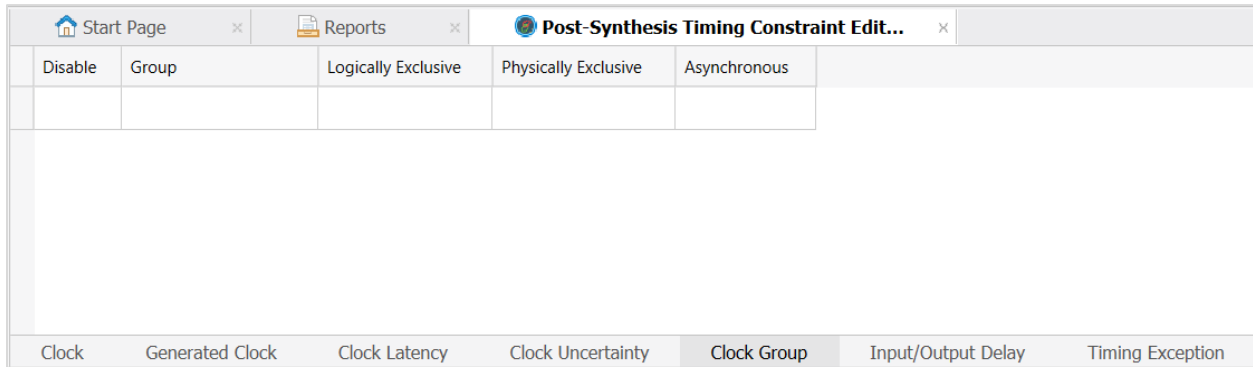


Figure 6.6 Post-Synthesis Timing Constraint Editor

- (3) Add constraint

[Figure 6.7](#) shows the setting of constraint. User should add group of ddr_sclk, instantiated_name_eclk and clk_50M respectively and set them as logically exclusive.

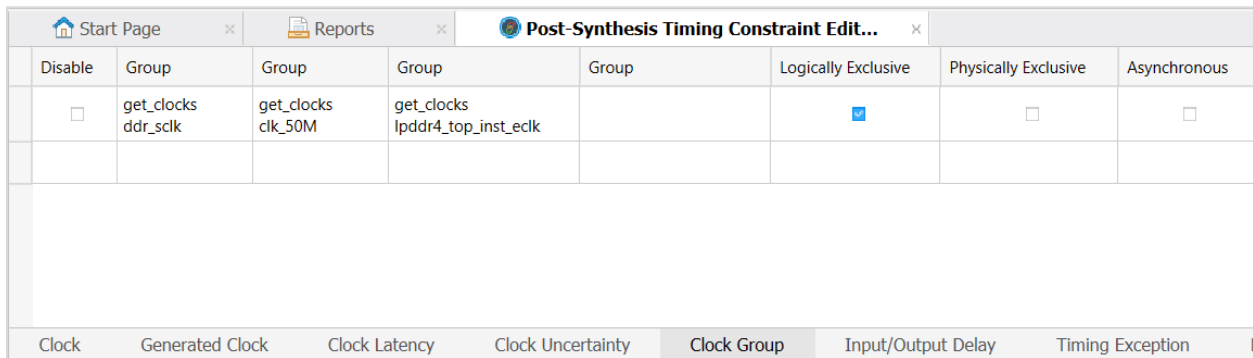


Figure 6.7 Setting of Constraint

- (4) Save constraint and continue to run the project

7. Example Design and Evaluation

Figure 7.1 shows the directory structure of LPDDR4 IP core. The 'lpddr4_ip' is the name of IP instantiated. And within this folder there are five subfolders, as described below.

- (1) 'constraints': there are fdc file and ldc file used for timing constraints but not include IO constraints.
- (2) 'eval': there is an evaluation software named 'LSCC_EMIF_EVAL.exe' and related RTL design used for hardware validation and evaluation.
- (3) 'misc': RTL instantiated template is provided.
- (4) 'rtl': IP source code and black box file
- (5) 'testbench': there are scripts provided for simulation in Modelsim and Questasim, named 'sim_modelsim.do' and 'sim_questasim.do' respectively. The .svp file is a LPDDR4 simulation model. Besides, the tb.v is also provided as the testbench.

There are 4 files named 'component.xml', 'design.xml', 'lpddr4_ip.cfg', 'lpddr4_ip.ipx' respectively. These files are IP configuration files, regenerating IP in Radiant will update these files.

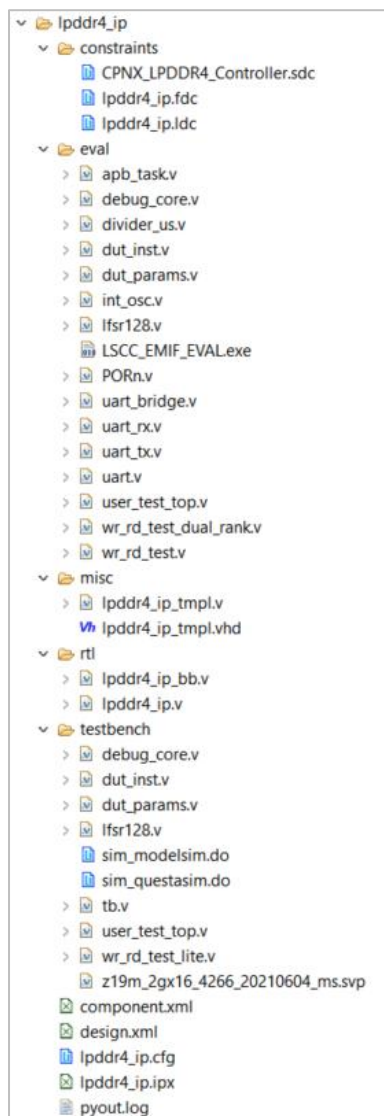


Figure 7.1 LPDDR4 Core Directory Structure

7.1. Simulation

Use script .do file simulation Steps:

- 0) As Figure 7.2 showed generate IP core with **Simulation Mode Enable** box checked and **Vref(CA/DQ) Training Enable** box unchecked
- 1) Open Modelsim or Questasim
- 2) cd {<IP path>\testbench}
- 3) do sim_modelsim.do or do sim_questasim.do

There are two types of simulation.

- (1) Conventional simulation

This configuration will perform all the training tasks. User needs to select the '**Simulation Mode Enable**' and all the other training tasks as Figure 7.2. Under this configuration, training time consumption could exceed 1ms (absolute time) in Questasim/Modelsim.

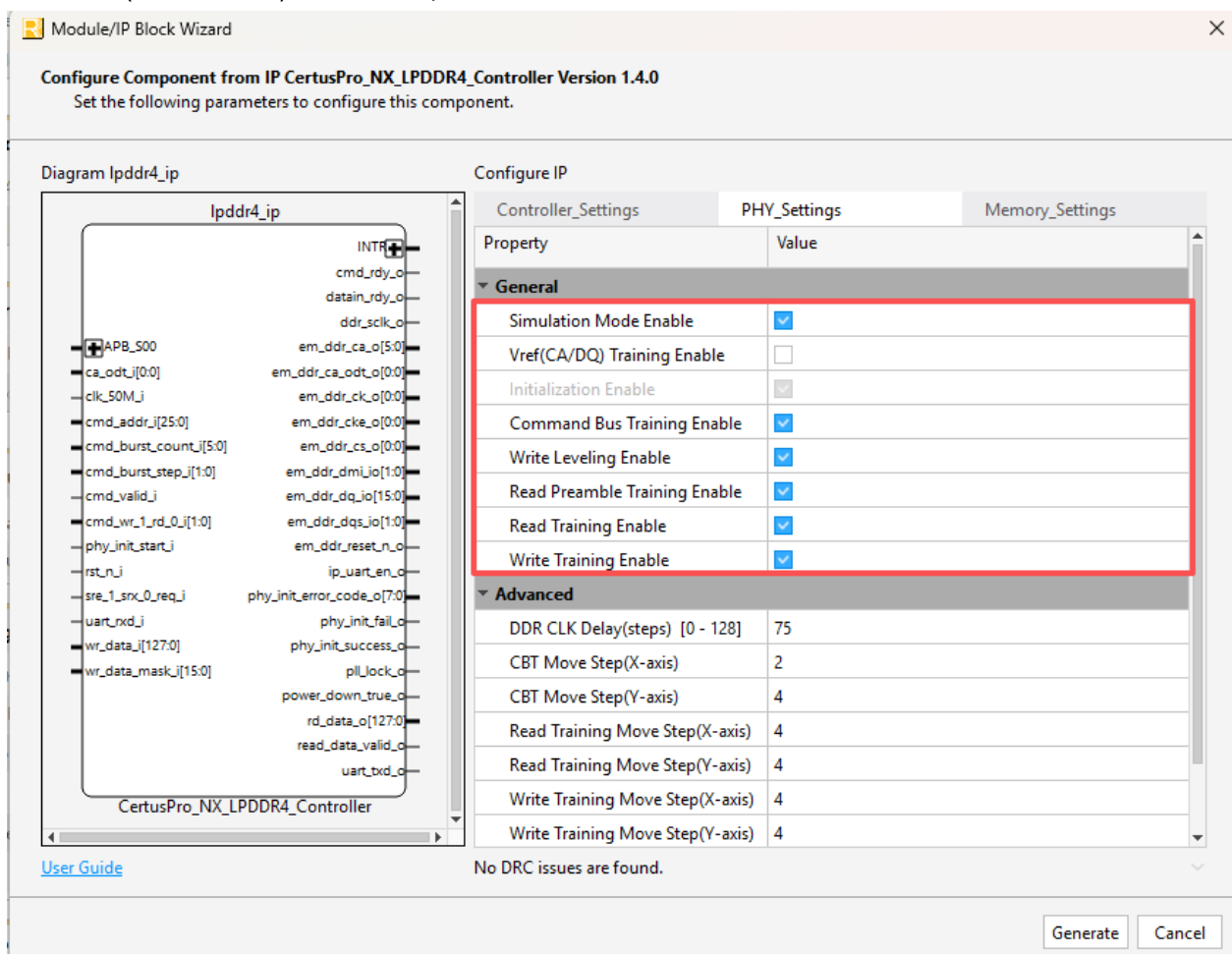


Figure 7.2 Normal Simulation Configuration

- (2) Accelerated simulation

This configuration uses pre-defined parameters to initialize the memory controller instead of training, which decreases the training time consumption significantly. User needs to deselect all the training steps as Figure 7.3. Under this configuration, training time consumption is about 750us (absolute time) in Questasim/Modelsim. For single rank simulation, the time consumption is about 450us (absolute time).

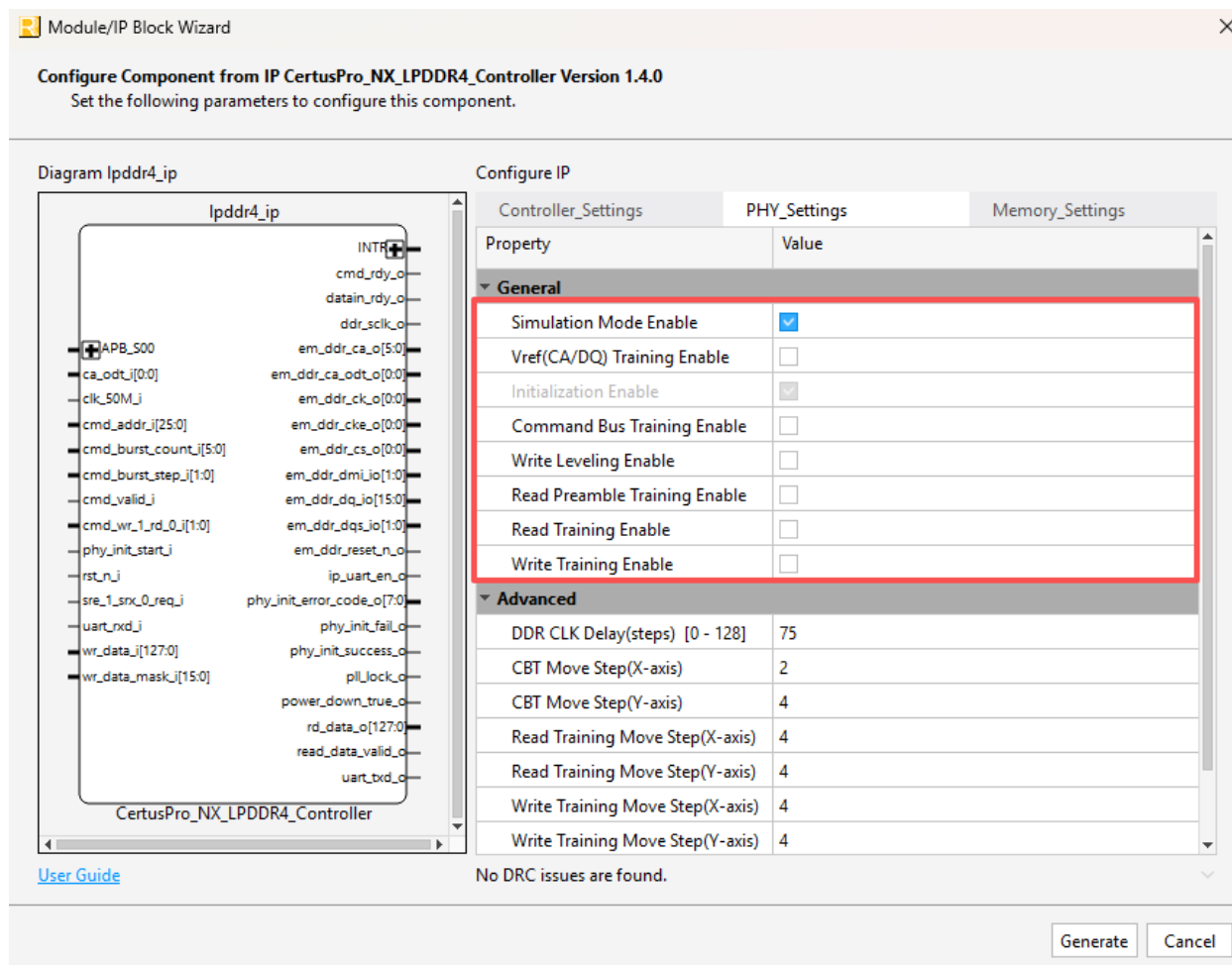


Figure 7.3 Accelerated Simulation Configuration

To be noticed that the simulation model “z19m_2gx16_4266_20210604_ms.svp” provided, is a 2Gb, single channel LPDDR4 SDRAM. Therefore, if this model is used, the maximum row width is 14. For a single rank design, the valid bit of input signal “cmd_addr_i” of this IP is [26:0], while for a dual rank design, the valid bit is [27:0].

7.2. Evaluation Design

The user can use the evaluation design source file under the directory of ‘lpddr4_ip/eval’ to build their own design. This design through UART RX select various test mode and UART TX feedback test result to user. So, make sure the UART is enabled on the hardware.

The Evaluation Design can be used for testing write/read functionality, efficiency under every burst count, write/read latency, reset cycle reliability, parameter tuning etc.

To be mentioned that, user should modify the parameters in user_test_top.v to adapt Memory data width and some other parameters. Besides, the LPDDR4 evaluation tool is also provided under the directory of ‘lpddr4_ip/eval’. More details please refer to [Evaluation Tool](#).

7.3. Evaluation Tool

The evaluation tool named ‘LSCC_EMIF_EVAL.exe’ is under the directory of ‘lpddr4_ip/eval’, for detail path, please refer to [Figure 7.1](#). The Latticesemi External Memory Interface Evaluation tool can be used for function test, parameter tuning and performance evaluate of external memory IP like DDR3, DDR4 and LPDDR4.

After successfully downloading Evaluation design and connecting UART cable, it is recommended to follow the operations below to explore how to start evaluation.

- 1) Open UART port, if use USB to UART through FTDI chip, usually there are two COM ports and the smaller number COM sometimes used for JTAG; larger number COM used for UART. As shown in [Figure 7.4](#) select COM4 and click Open Port
- 2) Select your target type in “Type Sel”, and click on “update”. This function configures the control panel and displays a message specific to the target memory.
- 3) Check the Displaying Training Detail box then click Reset and Re-Training button, the details of training can be found in the Message window, click Clear button will empty Message window.
- 4) According [Table 7.1](#) Select Burst count, Data Pattern, Test Mode, RD to WR Delay. Then click Start Test and observe Efficiency performance, data compare result, etc.
- 5) Auto Test tap also supports auto reset cycle test. Click Reset Cycle Start button will start reset cycle testing, to avoid too much message print suggest not displaying the training detail (uncheck the display training detail)
- 6) Clicking Reset Cycle Stop button can force stop testing or once initial fail or initial success times up to 8,388,607(24'h80_0000), will auto stop test. Click the Read Result button, the test result will show in the message window.
- 7) Using special test mode named 03-SR & Power Down test Mode can test self-refresh and power down function automatically, user also can click ‘Power Down & Self-Refresh Enter’ and ‘Power On & Self-Refresh Exit’ Buttons to test self-refresh function manually.

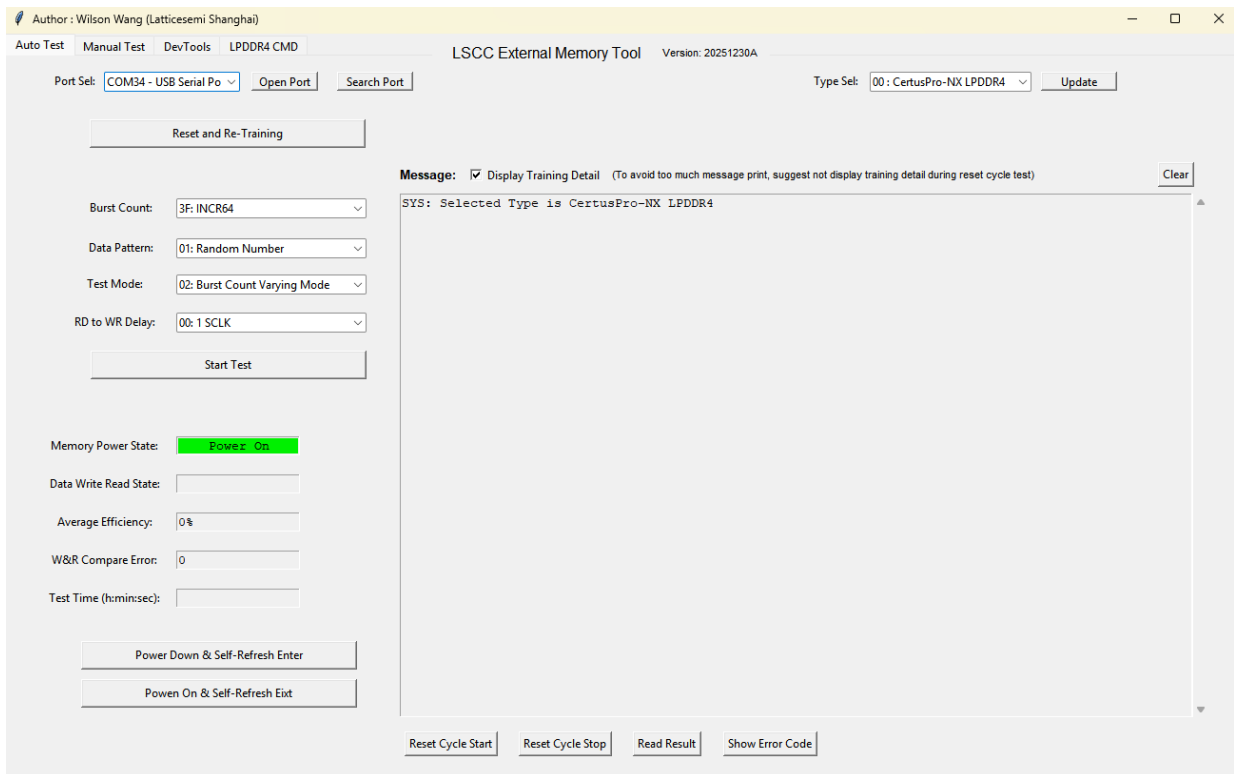


Figure 7.4 Auto Test Mode

Table 7.1 Test Mode Select

Items	Selectable Value	Description
Burst Count	INC1~INC64	This option controls cmd_burst_count_i
Data Pattern	Incremental Number;	wr_data_i is incremental number or random number. The incremental number is 8bit based, wider data is combination of repetitive 8bit incremental numbers; The random number is 128bit based, wider data is

	Random Number; Address Number	combination of repetitive 128bit incremental numbers; The address number is depending on user address, the write data is same as the address, for example write 1024 to address 1024.
Test Mode	00~07	00: Single WR to Single RD Mode; Alternating once writing command to once reading command 01: Multi WR to Multi RD Mode; Alternating multi writing commands to multi reading commands 02: Burst Count Varying Mode; The cmd_burst_count_i is varying, so the user burst count items setting will be ignored 03: SR & Power Down Mode; Test Self-refresh Enter->Power down->Power on->Self-refresh Exit -> Write -> Read -> Self-refresh Enter -> The data comparing should be correct, but efficiency is lower because the power down period does not allow any write/read. 04: Mask Write Mode; Used for testing write mask function, due to certain write bytes are masked, so the data comparing result is wrong. 05: Throughput Read Only Mode; Used for test read efficiency, in this mode, there is no write operation, so the data comparing is wrong. 06: Throughput Write Only Mode; Used for test write efficiency, in this mode, there is no read operation so there is no data compare. 07: Throughput WR + RD Mode; Used for test read write mix efficiency, in this mode, for maximum efficiency read operation maybe prior to write operation so the data compare maybe wrong.
RD to WR Delay	1sclk~256sclk	Read operation to Write operation delay, used for testing efficiency.

As shown in Figure 7.5 is manual test mode, the user can specify address, burst count, burst step. The write data will be auto generated. In the message window, will print read write data and latency (Calculate the time from command be accepted to first read/write data valid), Check the twice in a row box can eliminating active time and get a pure read/write latency.

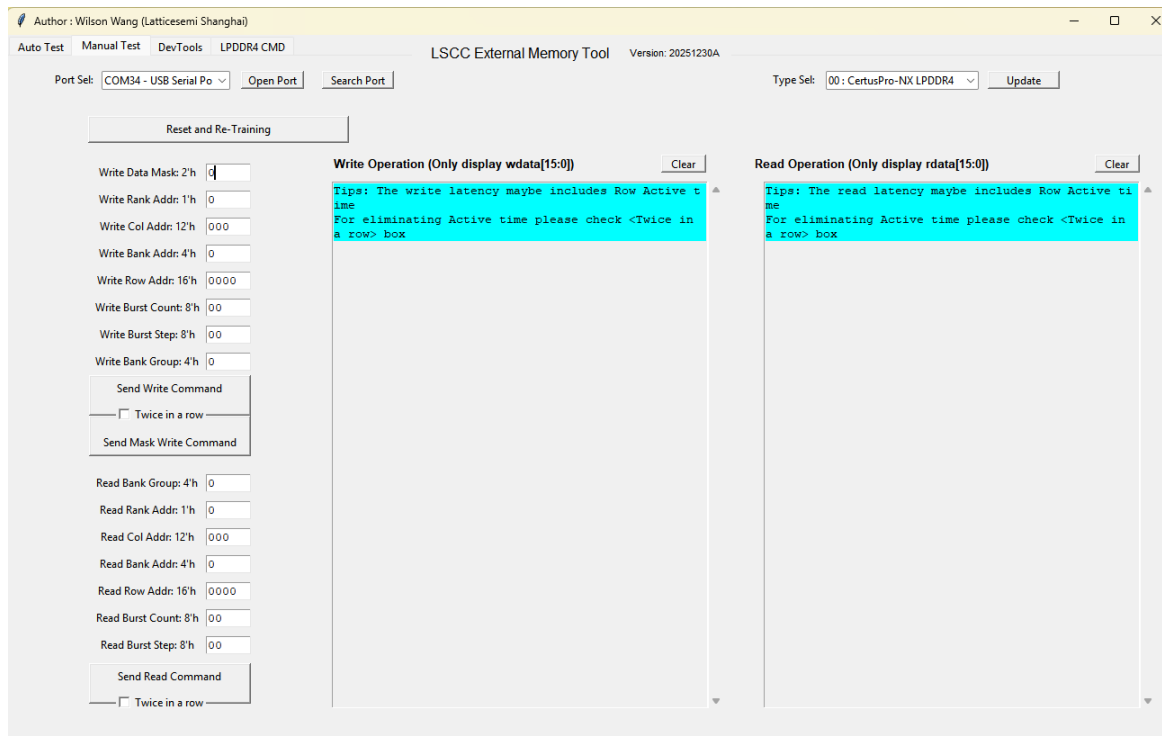


Figure 7.5 Manual Test Mode

As [Figure 7.6](#) shown is DevTools tap. The IP GUI configuration is before synthesizing, but during test and evaluation phase sometimes user wants to change some parameters immediately. This tap controls user APB BUS of IP CSR and can fast change parameters, detail please refer [Registers Description](#). This function is just used for evaluation and debugging.

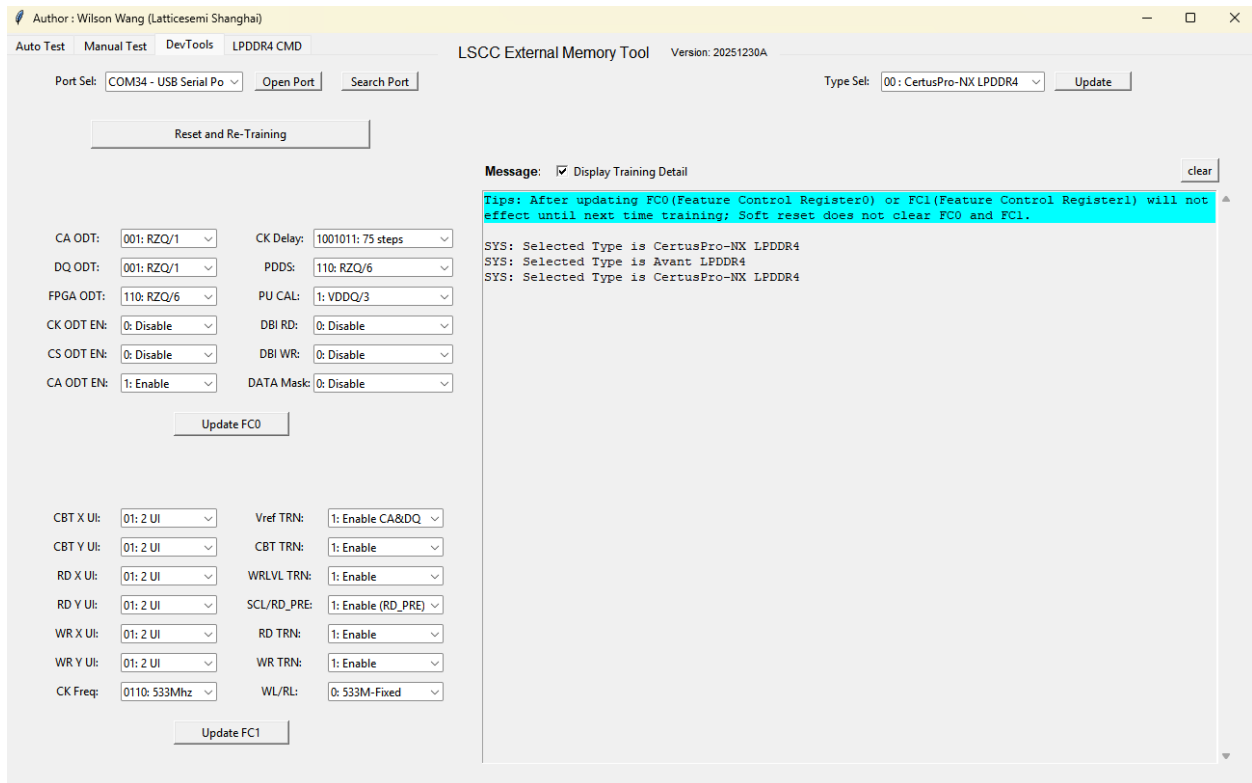


Figure 7.6 DevTools

As show in [Figure 7.7](#) CMD Analyze tap, it helps user fast analyze the LPDDR4 commands, look up the Mode register definition and VREF definition.

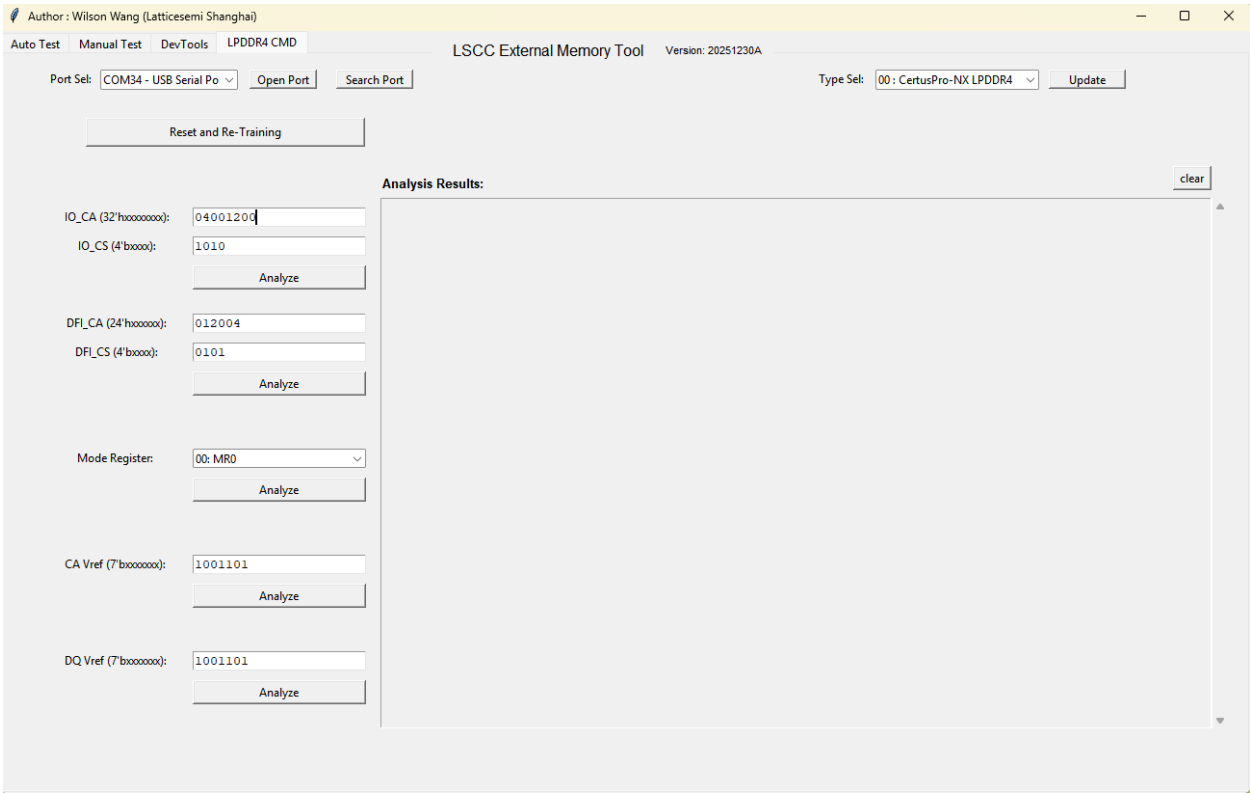


Figure 7.7 CMD Analyze

8. User Interface bridge reference design

8.1. Virtual FIFO

The Virtual FIFO design allowed user to access LPDDR4 MC like a DC-FIFO, and not care about the address. Users can specify the start address and end address of LPDDR4 to limit the FIFO storage range. Virtual FIFO design with MPMC can divide LPDDR4 to several independent FIFO.

As show in [Figure 8.1](#) This design provided write ports to input user data and read ports to output user data. For the convenience of the user, this design also processes cross clock domain put one DC-FIFO at LPDDR4 MC write side and another DC-FIFO at LPDDR4 read side, the two DC-FIFO bridge write clock or read clock to LPDDR4 MC SCLK. If 'wfifo_full' signal asserted; or 'rfifo_empty' asserted but 'unread_cnt' not zero, these two situations mean the LPDDR4 bandwidth is not enough. [Table 8.2](#) describes the ports of the Virtual FIFO design and [Table 8.3](#) describes the parameters of the Virtual FIFO design.

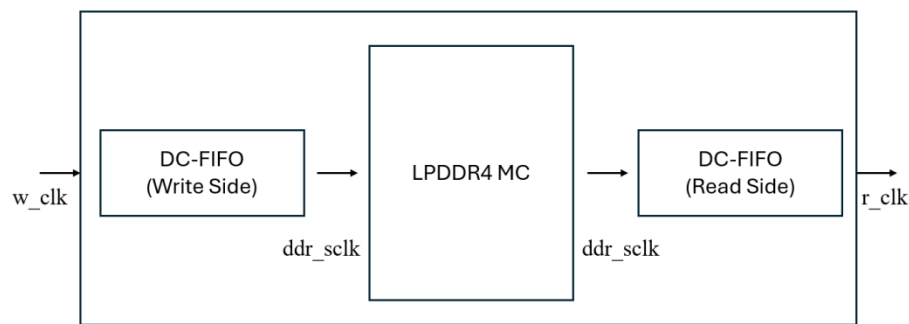


Figure 8.1 Virtual FIFO Diagram

8.1.1. Read Data Latency

The latency from `rd_en_i` to `q_o` depends on whether the output register is enabled in the read-side FIFO.

[Figure 8.2](#) shows the read timing with output register bypassed. When the output register is bypassed, the corresponding FIFO data appears at `q_o` one `rdclock_i` cycle after `rd_en_i` is asserted. This mode minimizes read latency.

[Figure 8.3](#) shows the read timing with output register enabled. When the output register is enabled, the corresponding FIFO data appears at `q_o` two `rdclock_i` cycles after `rd_en_i` is asserted. The additional output register stage adds one cycle of latency and can improve timing robustness.

For consecutive reads, `rd_en_i` may remain asserted for multiple `rdclock_i` cycles. After the initial latency, the FIFO outputs consecutive data words in order.

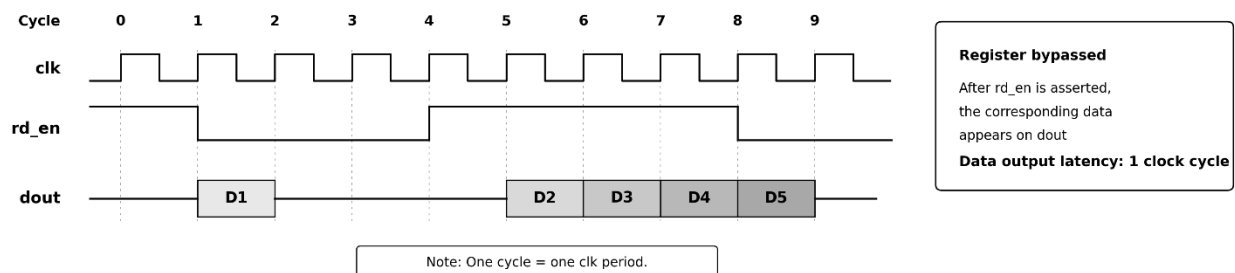


Figure 8.2 Read Timing with Output Register Bypassed

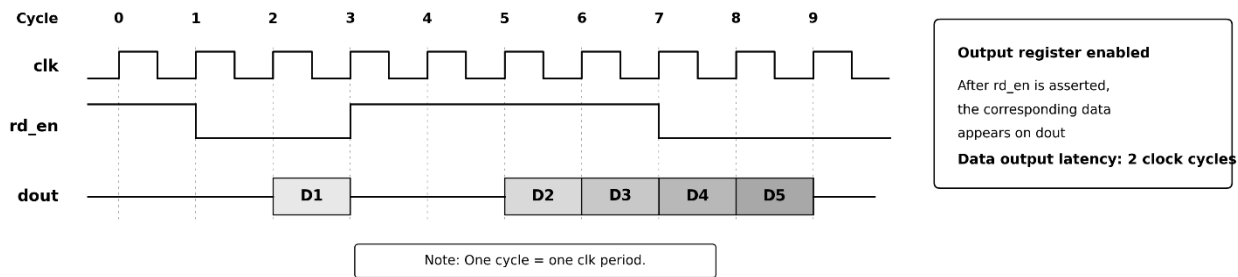


Figure 8.3 Read Timing with Output Register Enabled

8.1.2. Outstanding Command Counters

The Virtual FIFO tracks outstanding LPDDR4 write and read transactions with `wr_cmd_count_o` and `rd_cmd_count_o`. These counters are intended for status monitoring, debug, flow control, and for preventing reset or clear operations while a transaction is still in progress.

A write command handshake occurs on a rising edge of `ddr_sclk_i` when `cmd_valid_o`, `cmd_rdy_i`, and `cmd_wr_1_rd_0_o` are all high. The write-command counter increments at this handshake. After the write-data transfer completes and `datain_rdy_i` deasserts, the write-command counter decrements.

A read command handshake occurs on a rising edge of `ddr_sclk_i` when `cmd_valid_o` and `cmd_rdy_i` are high and `cmd_wr_1_rd_0_o` is low. The read-command counter increments at this handshake. After all data for the read command has been returned and `read_data_valid_i` deasserts, the read-command counter decrements.

The Virtual FIFO supports up to two outstanding write commands and up to two outstanding read commands. Therefore, the normal valid range of both `wr_cmd_count_o` and `rd_cmd_count_o` is 0 to 2.

When either `wr_cmd_count_o` or `rd_cmd_count_o` reaches 3, the user logic should treat this condition as an error. No additional command of the corresponding type should be issued until the outstanding-command count returns to a valid range.

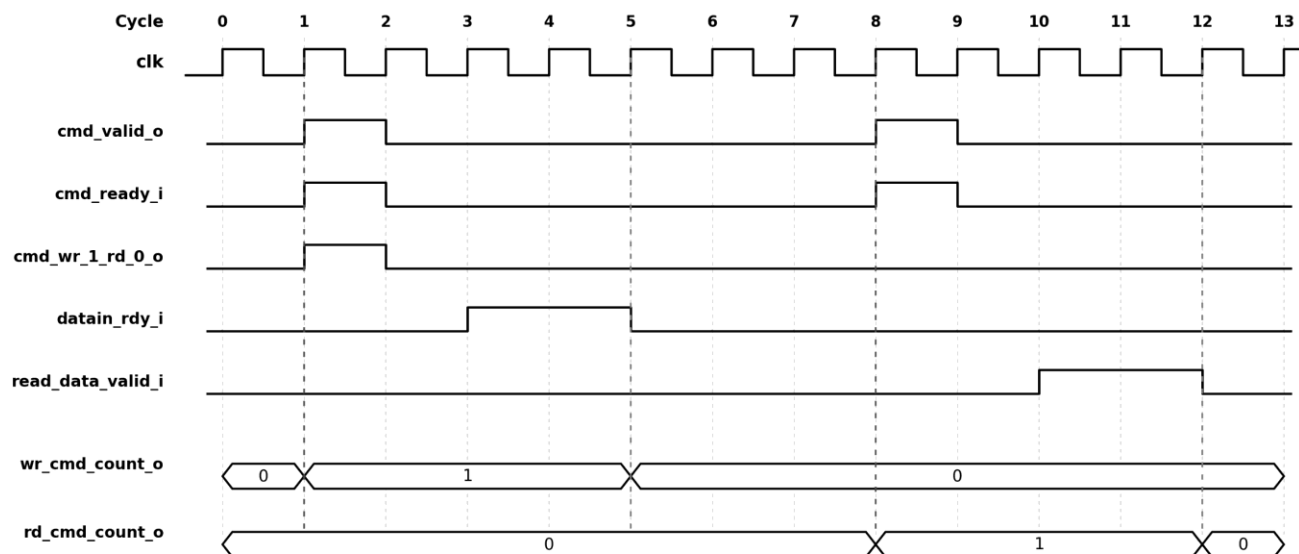


Figure 8.4 Outstanding Write and Read Command Counter Timing

Figure 8.4 shows an example sequence. All command, status, and counter transitions are aligned to `ddr_sclk_i` rising edges. In this example, `datain_rdy_i` rises two clock cycles after the write handshake; after `datain_rdy_i` falls, two full idle clock cycles elapse before the read command is issued.

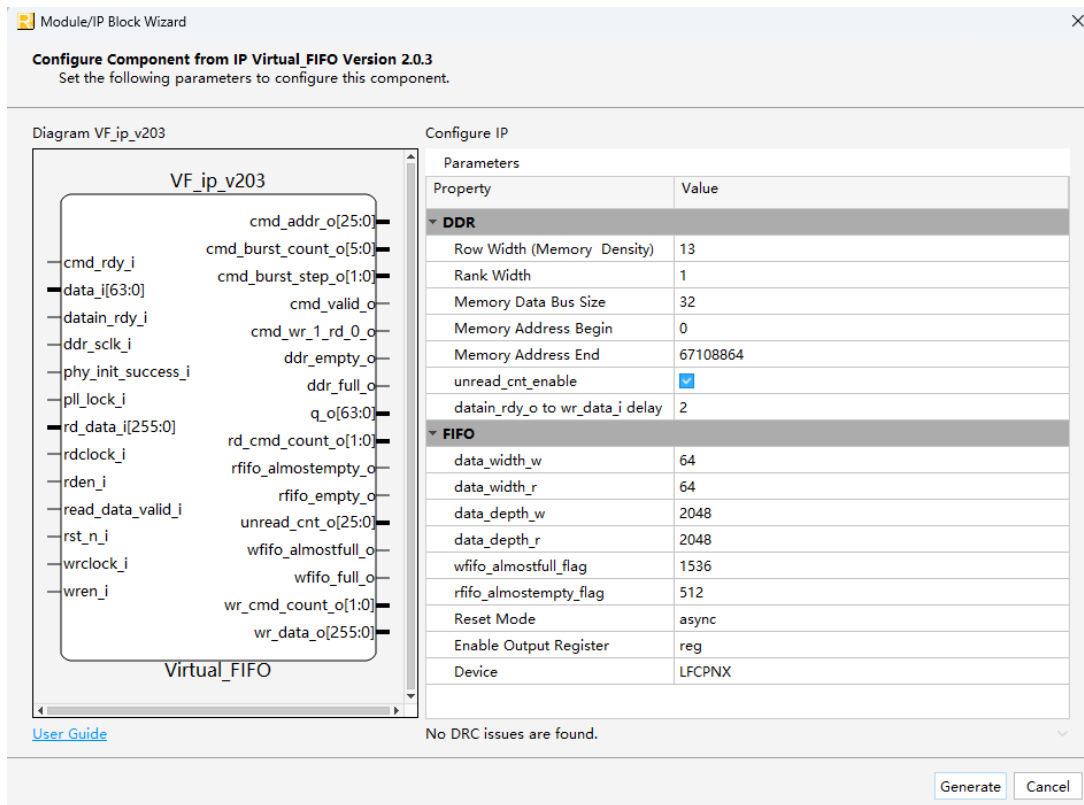


Figure 8.5 Virtual FIFO Configuration GUI

Table 8.1 provides quick facts about CertusPro-NX LPDDR4 SDRAM Controller IP.

Table 8.1 Virtual FIFO Design Quick Fact

Support Device	CertusPro-NX Family	
Resource Utilization	LUT4s	485
	EBR	8 (Base two 128x512 DC FIFOs)
Performance	Efficiency	82%

Table 8.2 Virtual FIFO Design Ports

Port name	I/O	Width	Description
Port Connect with CPNX LPDDR4 Memory Controller IP			
ddr_sclk_i	input	1	Please refer Table 4.1
pll_lock_i	input	1	
init_success_i	input	1	
cmd_rdy_i	input	1	
cmd_valid_o	output	1	
cmd_address_o	output	Row Width + 3 + 10 (Single RANK); Row Width+ 1 + 3 + 10 (Dual RANK)	
cmd_burst_count_o	output	6	

cmd_burst_step_o	output	2	
cmd_wr_1_rd_0_o	output	1	
datain_rdy_i	input	1	
wr_data_o	output	LPDDR4 DQ width*8	
rd_data_i	input	LPDDR4 DQ width*8	
read_data_valid_i	input	1	
Write Interface Port			
wrclock_i	input	1	User write data clock and will be connect to DC-FIFO wclk
wren_i	input	1	User write data valid and will be connect to DC-FIFO we
data_i	input	2^n	User write data and will be connect to DC-FIFO wdata
wfifo_full_o	output	1	Write side DC-FIFO is full it means LPDDR4 MC peak bandwidth is not enough
wfifo_almostfull_o	output	1	Write side DC-FIFO is almost full
Read Interface Port			
rdclock_i	input	1	User read data clock and will be connect to DC-FIFO rclk
rden_i	input	1	User read data valid and will be connect to DC-FIFO re
q_o	output	2^n	User read data and will be connect to DC-FIFO rdata
rfifo_empty_o	output	1	Read side DC-FIFO is empty it means LPDDR4 MC peak bandwidth is not enough, or there is no unread data stored in LPDDR4 Memory
rfifo_almostempty_o	output	1	Read side DC-FIFO is almost empty
FIFO Counter (Optional)			
unread_cnt	output	Row Width + 3 + 10 (Single RANK); Row Width+ 1 + 3 + 10 (Dual RANK)	Indicate how many unread data stored in LPDDR4 Memory, the unit is depends on p_ddr_dq_width; for example if p_ddr_dq_width=16, and unread_cnt is 1 it means there is 1*16/8=2 byte; if p_ddr_dq_width=32, and unread_cnt is 4 it means there is 4*32/8=16 byte; It is LPDDR4 sclk domain This port only available when 'unread_cnt Enable' be checked in GUI
ddr_empty_o	output	1	Indicate LPDDR4 is empty, it is LPDDR4 sclk domain This port only available when 'unread_cnt Enable' be checked in GUI
ddr_full_o	output	1	Indicate LPDDR4 is full, it is LPDDR4 sclk domain This port only available when 'unread_cnt Enable' be checked in GUI
wr_cmd_count_o	output	2	A write command has already been issued to the LPDDR memory controller, but the write-data transfer has not yet completed.
rd_cmd_count_o	output	2	A read command has already been issued to the LPDDR memory controller, but not all read data has been received yet.
Reset			
rst_n_i	input	1	Asynchronous reset, active low

Table 8.3 Virtual FIFO Design Parameters

Attribute	Selectable Values	Default	Dependency on Other Attributes
LPDDR4			
Row Width (Memory Density)	13 (Dual Channel-2Gb), 14 (Dual Channel-4Gb), 15 (Dual Channel-8Gb), 16 (Dual Channel-16Gb), 17 (Dual Channel-32Gb)	13	Dual Channel 32Gbit: Row width is 17 Dual Channel 24Gbit: Row width is 17 Dual Channel 16Gbit: Row width is 16 Dual Channel 12Gbit: Row width is 16 Dual Channel 8Gbit: Row width is 15 Dual Channel 6Gbit: Row width is 15 Dual Channel 4Gbit: Row width is 14 Dual Channel 2Gbit: Row width is 13
Rank Width	1, 2	1	Rank number, 1 is single rank, 2 is dual rank
Memory Data Bus Size	16, 32, 64	32	16bit will occupy 1 FPGA HP bank 32bit will occupy 2 FPGA HP bank 64bit will occupy 3 FPGA HP bank
Memory Address Begin	x-<Max that can fit in the device>	67108864	Virtual FIFO write/read starting address of LPDDR4
Memory Address End	x-<Max that can fit in the device>	0	Virtual FIFO write/read end address of LPDDR4
datain_rdy_o to wr_data_i delay	1, 2	2	When IP core assert datain_rdy_o need wait 1 or 2 sclk to input write data
unread_cnt_enable	Enable, Disable	Disable	Enable Stream FIFO counter, used to indicate how many data stored in the FIFO, it is LPDDR4 sclk domain
FIFO			
data_width_w	32, 64, 128, 256, 512,	64	User write data width
data_width_r	32, 64, 128, 256, 512,	64	User Read data width
data_depth_w	x-<Max that can fit in the device>	2048	To perform continuous read/write operations with LPDDR4, the depth of the FIFO must be greater than twice the size of burst read/write operations. The constraint is given by $y = x * \text{data_depth_w} / (\text{Memory Data Bus Size} * 8)$, where y must be at least 512. Additionally, since LPDDR4 requires time for self-refresh, a greater depth leads to higher efficiency. The default setting for y is 1024, at which it achieves 82% effective bandwidth.
data_depth_r	$\text{data_width_w} * \text{data_depth_w} / \text{data_width_r}$	2048	Decide by other data width and depth.
wfifo_almostfull_flag	1-<Max that can fit in the device>	1536	Less than write FIFO Depth.
rfifo_almostempty_flag	1-<Max that can fit in the device>	512	Less than read FIFO Depth.

8.2. AXI-4 I/F

The AXI-4 interface design converts user AXI-4 logic to LPDDR4 localbus, it enabled user access LPDDR4 IP as an AXI-4 target, but it will add extra access delay than access localbus directly. [Figure 8.6](#) shows the AXI-4 design configuration GUI.

- Support AXI-4 General 5 channels
 - Read address channel
 - Read data channel
 - Write address channel
 - Write data channel
 - Write response channel
- For the AXI-4 burst type, only support "INCR"
- For the AXI-4 address width, support 27, 28, 29, 30, 31, 32, 33, 34, match with address width of LPDDR4
- For the AXI-4 data width, support 32, 64, 128, 256
- For the LPDDR4 data width, support 128 (16bit DQ), 256bits (32bit DQ)
- For the AXI-4 AxLEN width, support 8
- Support the LPDDR4 Mask Write Function
- Support AXI-4 clock and LPDDR4 sclk cross clock domain

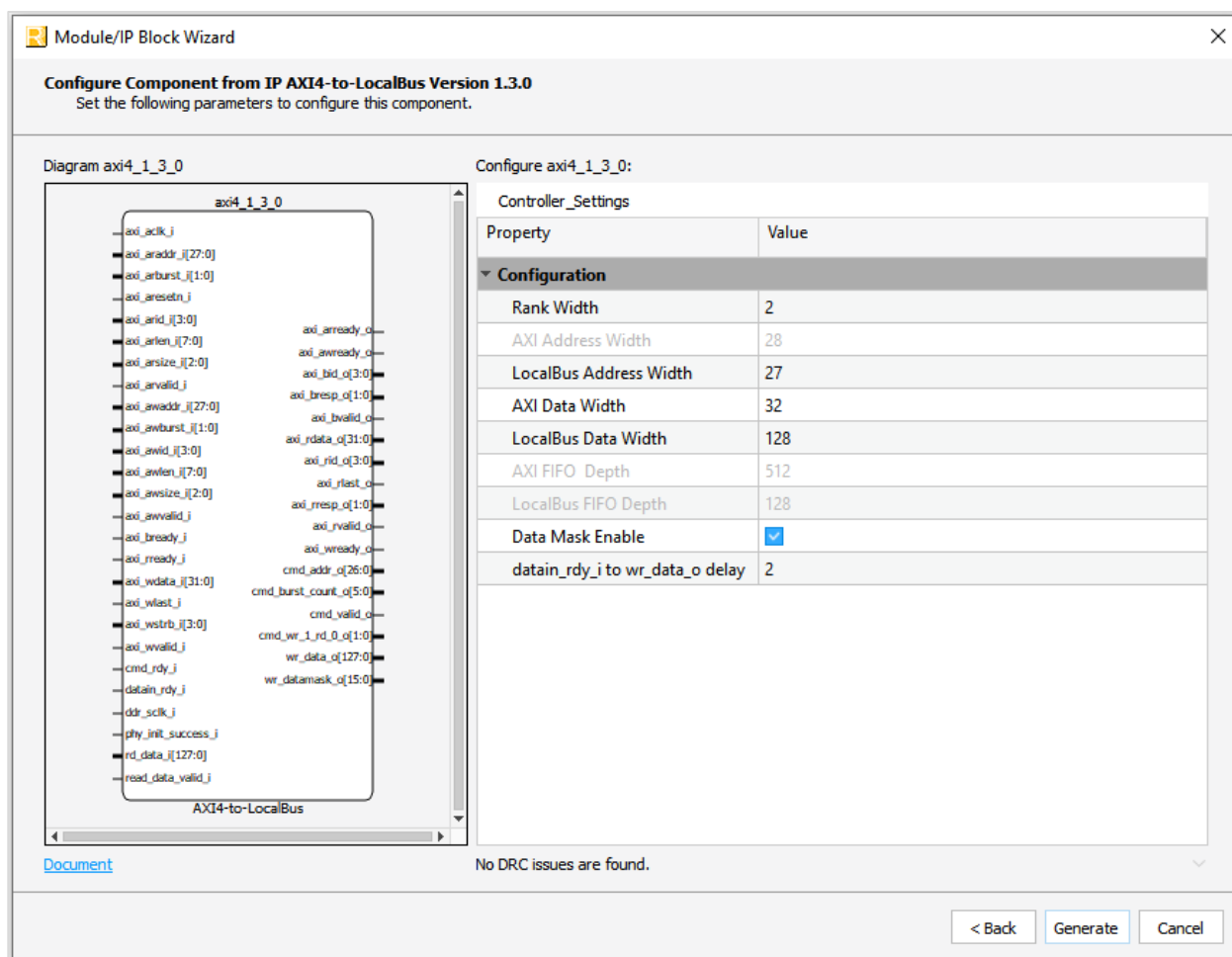


Figure 8.6 AXI-4 Configuration GUI

Table 8.4 is the AXI I/F design resource usage and Table 8.5 is the AXI I/F design performance, the performance test is based $\text{ddr_sclk_i} = \text{axi_aclk_i}$, so when AXI data width smaller than localbus width the efficiency even larger than 87%, it doesn't mean LPDDR4 efficiency can up to that, and it just AXI-4 side efficiency.

Table 8.4 AXI I/F Resource

AXI_DATA_WIDTH	LocalBus_DATA_WIDTH	LUT4	Registers	EBR
32	128	573	648	9
32	256	570	793	17
64	128	615	775	9
64	256	657	921	17
128	128	791	1058	9
128	256	744	1180	17
256	128	1163	1578	17
256	256	1151	1723	17

Table 8.5 AXI I/F Performance

AXI_DATA_WIDTH	LocalBus_DATA_WIDTH	LEN	Efficiency	Write Latency (AXI_awvalid & AXI_arvalid & AXI_arready) to AXI_rvalid
----------------	---------------------	-----	------------	--

				AXI_awready) to AXI_wvalid)	
32	128	16	72%	2 axi clock	32 axi clock
		32	81%	2 axi clock	32 axi clock
		64	87%	2 axi clock	32 axi clock
		128	92%	2 axi clock	32 axi clock
		256	97%	2 axi clock	32 axi clock
32	256	16	72%	2 axi clock	32 axi clock
		32	81%	2 axi clock	32 axi clock
		64	87%	2 axi clock	32 axi clock
		128	92%	2 axi clock	32 axi clock
		256	97%	2 axi clock	32 axi clock
64	128	16	72%	2 axi clock	32 axi clock
		32	81%	2 axi clock	32 axi clock
		64	87%	2 axi clock	32 axi clock
		128	92%	2 axi clock	32 axi clock
		256	97%	2 axi clock	32 axi clock
64	256	16	72%	2 axi clock	32 axi clock
		32	81%	2 axi clock	32 axi clock
		64	87%	2 axi clock	32 axi clock
		128	92%	2 axi clock	32 axi clock
		256	97%	2 axi clock	32 axi clock
128	128	16	62%	2 axi clock	32 axi clock
		32	68%	2 axi clock	32 axi clock
		64	72%	2 axi clock	32 axi clock
		128	82%	2 axi clock	32 axi clock
		256	82%	2 axi clock	32 axi clock
128	256	16	72%	2 axi clock	32 axi clock
		32	81%	2 axi clock	32 axi clock
		64	87%	2 axi clock	32 axi clock
		128	92%	2 axi clock	32 axi clock
		256	97%	2 axi clock	32 axi clock
256	128	16	36%	2 axi clock	32 axi clock
		32	38%	2 axi clock	32 axi clock
		64	39%	2 axi clock	32 axi clock
		128	41%	2 axi clock	32 axi clock
		256	41%	2 axi clock	32 axi clock
256	256	16	62%	2 axi clock	32 axi clock
		32	68%	2 axi clock	32 axi clock
		64	72%	2 axi clock	32 axi clock
		128	82%	2 axi clock	32 axi clock
		256	82%	2 axi clock	32 axi clock

Table 8.6 describes the port of AXI-4 I/F design, and Table 8.7 describes the parameters of AXI-4 I/F design.

Table 8.6 The port of AXI-4 I/F

Port Name	I/O	Width	Description
Clock and Reset			
axi_aclk_i	I	1	The AXI-4 clock signal.
ddr_sclk_i	I	1	The LPDDR4 clock. This is ¼ of the DDR clock frequency and is the main clock of the LPDDR4 Memory Controller IP.
axi_aresetn_i	I	1	Asynchronous active low reset.
AXI-4 Interface			
axi_arid_i	I	4	Read address ID
axi_araddr_i	I	AXI_ADDR_WIDTH	Read address
axi_arlen_i	I	8	Read burst count, 0-255 mean the len is 1-256
axi_arburst_i	I	2	Read burst type, only support "INCR"
axi_arvalid_i	I	1	Read address valid
axi_arready_o	O	1	Read address ready
axi_rid_o	O	4	Read ID, the value of the RID needs to match the value of ARID.
axi_rdata_o	O	AXI_DATA_WIDTH	Read data
axi_rresp_o	O	2	Read response
axi_rlast_o	O	1	The last data transmitted by the read command
axi_rvalid_o	O	1	Read data valid
axi_rready_i	I	1	Read data ready
axi_awid_i	I	4	Write address ID
axi_awaddr_i	I	AXI_ADDR_WIDTH	Write address
axi_awlen_i	I	8	Write burst count, 0-255 mean the len is 1-256
axi_awburst_i	I	2	Write burst type, only support "INCR"
axi_awvalid_i	I	1	Write address valid
axi_awready_o	O	1	Write address ready
axi_wdata_i	I	AXI_DATA_WIDTH	Write data
axi_wstrb_i	I	AXI_DATA_WIDTH/8	The byte qualifier that indicates whether the content of the associated byte of axi_wdata_i is processed as a data byte or a position byte.
axi_wlast_i	I	1	The last data transmitted by the write command
axi_wvalid_i	I	1	Write data valid
axi_wready_o	O	1	Write data ready
axi_bid_o	O	4	Response ID, the value of the BID needs to match the value of AWID
axi_bresp_o	O	2	Write response
axi_bvalid_o	O	1	Write response valid
axi_bready_i	I	1	Write response ready
LPDDR4 Interface			
phy_init_success_i	I	1	LPDDR4 initialization and Calibration success, user can write read memory
cmd_valid_o	O	1	LPDDR4 command and address valid output. When asserted, the addr, cmd, ofly_burst_len and cmd_burst_cnt

			inputs are considered valid.
cmd_rdy_i	I	1	LPDDR4 command ready input – When asserted, indicates that the core is ready to accept the next command and the corresponding address.
cmd_wr_1_rd_0_o	O	1 or 2	LPDDR4 command output to the memory controller.
cmd_addr_o	O	LPDDR_ADDR_WIDTH	LPDDR4 read or write address output to the memory controller.
cmd_burst_count_o	O	6	LPDDR4 command burst count output – Indicates the number of times a given read or write command is to be repeated by the controller automatically. 0 means one BL16, 1 means two BL16, 63 means 64 BL16 .
datain_rdy_i	I	1	Data ready input – When asserted, indicates the core is ready to receive the write data. There is 1 or 2 sclk delay between datain_rdy_o to wr_data_i
wr_data_o	O	LPDDR4_DATA_WIDTH	Write data output from AXI-4 to the memory controller. The write data width is x8 times the memory databus.
wr_datamask_o	O	LPDDR4_DATA_WIDTH/8	Write datamask output from AXI-4 to the memory controller.
read_data_valid_i	I	1	Read data valid output – When asserted, indicates the data on the read_data bus is valid.
rd_data_i	I	LPDDR4_DATA_WIDTH	Read data output from memory controller to the AXI-4.

Table 8.7 The parameters of AIX-4 I/F

Attribute	Selectable Value	Default	Description
Rank Width	1,2	2	The Rank number
AXI Address Width	27, 28, 29, 30, 31, 32, 33, 34	28	The AXI-4 address width
LocalBus Address Width	26, 27, 28, 29, 30, 31	27	The LPDDR4 address width
AXI Data Width	32, 64, 128, 256	32	The AXI-4 data width
LocalBus Data Width	128, 256	128	The LPDDR4 data width
AXI FIFO Depth	256,512,1024	512	The AXI FIFO Depth
LocalBus FIFO Depth	128,256,512	128	The LocalBus FIFO Depth
Data Mask Enable	“Enable”, “Disable”	“Enable”	The data mask function enable
datain_rdy_i to wr_data_o delay	1,2	2	Match the LPDDR4 IP

Table 8.8 The address width setting for LocalBus Address Width and AXI Address Width

Configure	Address Width					
LocalBus Address Width	26	27	28	29	30	31
AXI Address Width (for LPDDR4 x16 Single Rank)	27	28	29	30	31	--
AXI Address Width (for LPDDR4 x16 Dual Rank)	--	29	30	31	32	33
AXI Address Width (for LPDDR4 x32 Single Rank)	28	29	30	31	32	--
AXI Address Width (for LPDDR4 x32 Dual Rank)	--	30	31	32	33	34

Table 8.8 describes address width setting for LocalBus Address Width and AXI Address Width.

Table 8.9 The AxLEN setting

LocalBus_DATA_WIDTH \ AXI_DATA_WIDTH		AxLEN			
		32bit	64bit	128bit	256bit
Burst count	128bit	1:8	1:4	1:2	1:1
	256bit	1:16	1:8	1:4	1:2

Table 8.9 describes the AxLEN setting of AXI-4 I/F design.

Below are some usage limitations

- The AxLEN * AXI_DATA_WIDTH can't smaller than LocalBus_Burst_Count * LocalBus_DATA_WIDTH*2; For example, AXI_DATA_WIDTH is 32, LocalBus_DATA_WIDTH is 128, the minimum AxLEN is 8.
- For LPDDR4 Memory data path widths of x16, start address aligned with 32n
- For LPDDR4 Memory data path widths of x32, start address aligned with 64n

8.3. Avalon I/F

The Avalon interface design converts user Avalon logic to LPDDR4 localbus, it enabled user access LPDDR4 IP as an Avalon target but will add extra access delay than localbus. Figure 8.7 shows the Avalon design configuration GUI.

- Support Write and Read Burst for Avalon
- For the Avalon address width, support 27, 28, 29, 30, 31, 32, 33, 34, match with address width of LPDDR4
- For the Avalon data width, support 32, 64, 128, 256
- For the LPDDR4 data width, support 128, 256
- For the Avalon burst count width, support 8
- For the LPDDR4 burst count, support 1-64
- Support the LPDDR4 Mask Write Function
- Support Avalon and LPDDR cross clock domain

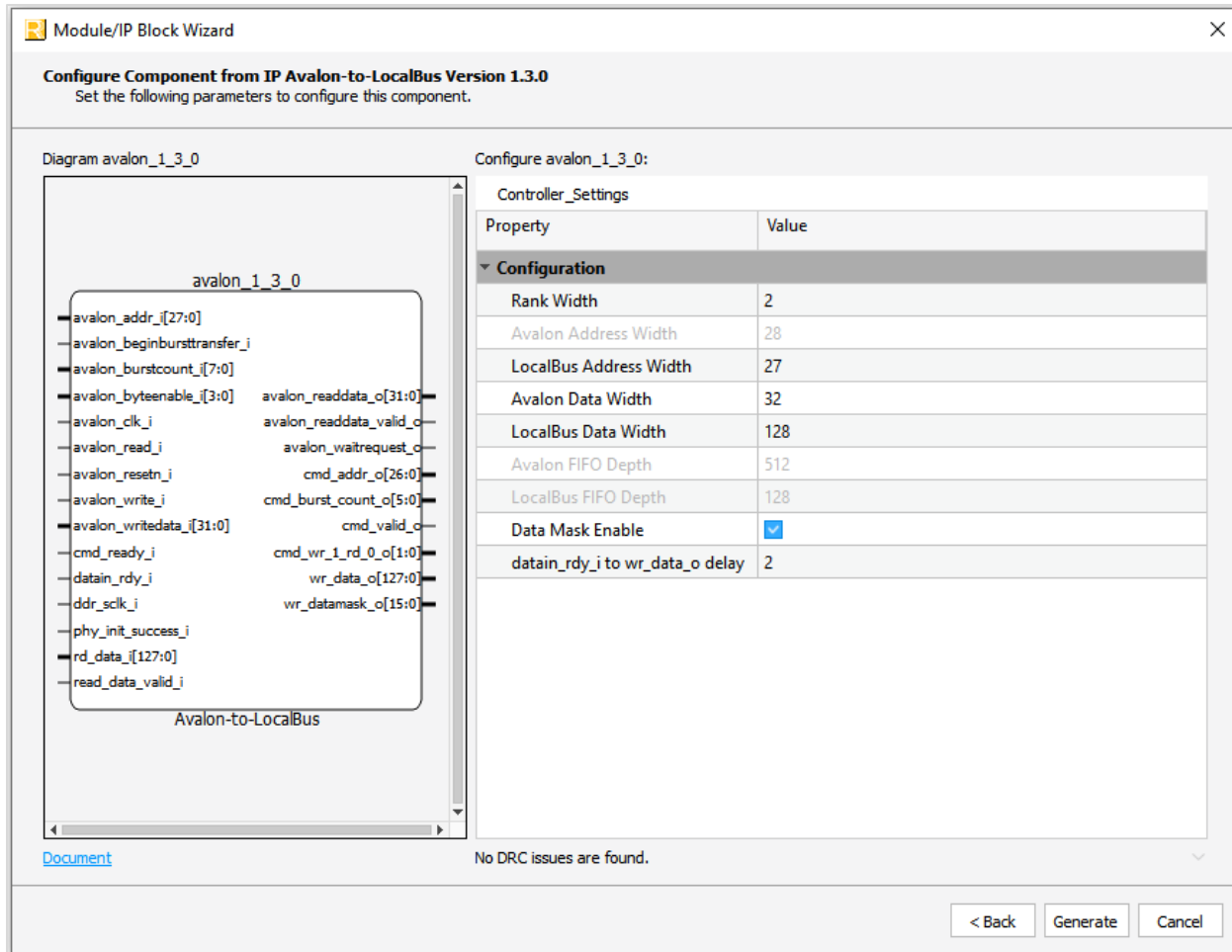


Figure 8.7 Avalon Configuration GUI

Table 8.10 is the Avalon I/F design resource usage and Table 8.11 is the Avalon I/F design performance, the performance test is based $\text{ddr_sclk_i} = \text{avalon_clk_i}$, so when Avalon data width smaller than localbus width the efficiency even larger than 87%, it doesn't mean LPDDR4 efficiency can up to that, and it just Avalon side efficiency.

Table 8.10 Resource of Avalon I/F

Avalon_DATA_WIDTH	LocalBus_DATA_WIDTH	LUT4	Registers	EBR
32	128	504	641	9
32	256	517	783	17
64	128	508	672	9
64	256	522	815	17
128	128	525	757	9
128	256	503	877	17
256	128	552	889	17
256	256	537	1027	17

Table 8.11 Performance of Avalon I/F

Avalon_DATA_WIDTH	LocalBus_DATA_WIDTH	LEN	Efficiency	Write Latency (avalon_write_i & avalon_beginbursttransf er_i) to	Read Latency (avalon_read_i & avalon_beginbursttransf er_i) to
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				er_i) to (avalon_waitrequest_o)	avalon_readdata_valid_ o
32	128	16	36%	1 avalon clock	32 avalon clock
		32	53%	1 avalon clock	32 avalon clock
		64	68%	1 avalon clock	32 avalon clock
		128	79%	1 avalon clock	32 avalon clock
		256	90%	1 avalon clock	32 avalon clock
32	256	16	36%	1 avalon clock	32 avalon clock
		32	53%	1 avalon clock	32 avalon clock
		64	68%	1 avalon clock	32 avalon clock
		128	79%	1 avalon clock	32 avalon clock
		256	90%	1 avalon clock	32 avalon clock
64	128	16	36%	1 avalon clock	32 avalon clock
		32	53%	1 avalon clock	32 avalon clock
		64	68%	1 avalon clock	32 avalon clock
		128	79%	1 avalon clock	32 avalon clock
		256	90%	1 avalon clock	32 avalon clock
64	256	16	36%	1 avalon clock	32 avalon clock
		32	53%	1 avalon clock	32 avalon clock
		64	68%	1 avalon clock	32 avalon clock
		128	79%	1 avalon clock	32 avalon clock
		256	90%	1 avalon clock	32 avalon clock
128	128	16	36%	1 avalon clock	32 avalon clock
		32	52%	1 avalon clock	32 avalon clock
		64	68%	1 avalon clock	32 avalon clock
		128	79%	1 avalon clock	32 avalon clock
		256	84%	1 avalon clock	32 avalon clock
128	256	16	36%	1 avalon clock	32 avalon clock
		32	53%	1 avalon clock	32 avalon clock
		64	68%	1 avalon clock	32 avalon clock
		128	79%	1 avalon clock	32 avalon clock
		256	90%	1 avalon clock	32 avalon clock
256	128	16	36%	1 avalon clock	32 avalon clock
		32	43%	1 avalon clock	32 avalon clock
		64	45%	1 avalon clock	32 avalon clock
		128	45%	1 avalon clock	32 avalon clock
		256	43%	1 avalon clock	32 avalon clock
256	256	16	36%	1 avalon clock	32 avalon clock
		32	52%	1 avalon clock	32 avalon clock
		64	68%	1 avalon clock	32 avalon clock
		128	79%	1 avalon clock	32 avalon clock
		256	84%	1 avalon clock	32 avalon clock

Table 8.12 describes the port of Avalon I/F design, and Table 8.13 describes the parameters of Avalon I/F design.

Table 8.12 The Interfaces Description of Avalon to LocalBus

Port Name	I/O	Width	Description
Clock and Reset			
avalon_clk_i	I	1	The Avalon clock signal.
ddr_sclk_i	I	1	The LPDDR4 clock. This is ¼ of the DDR clock frequency and is the main clock of the LPDDR4 Memory Controller IP.
avalon_resetsn_i	I	1	Asynchronous active low reset.
Avalon Interface			
avalon_addr_i	I	Avalon_ADDR_WIDTH	The address of Avalon
avalon_byteenable_i	I	Avalon_DATA_WIDTH/8	Enables one or more specific byte channels during transmission on an interface.
avalon_read_i	I	1	Read burst is enable
avalon_readdata_o	O	Avalon_DATA_WIDTH	Read data
avalon_readdatavalid_o	O	1	Read data valid
avalon_write_i	I	1	Write burst is enable
avalon_writedata_i	I	Avalon_DATA_WIDTH	Write data
avalon_waitrequest_o	O	1	Avalon_waitrequest_o is set when it cannot respond to read or write requests.
avalon_burstcount_i	I	8	Indicates the number of transmissions in each burst. 0-255 means the burst count is 1-256
avalon_beginbursttransfer_i	I	1	The first cycle of a burst is set to indicate the start of a burst transmission.
LPDDR4 Interface			
phy_init_success_i	I	1	LPDDR4 initialization and Calibration success, user can write read memory
cmd_valid_o	O	1	LPDDR4 command and address valid output. When asserted, the addr, cmd, ofly_burst_len and cmd_burst_cnt inputs are considered valid.
cmd_rdy_i	I	1	LPDDR4 command ready input – When asserted, indicates that the core is ready to accept the next command and the corresponding address.
cmd_wr_1_rd_0_o	O	1	LPDDR4 command output to the memory controller.
cmd_addr_o	O	LPDDR_ADDR_WIDTH	LPDDR4 read or write address output to the memory controller.
cmd_burst_count_o	O	6	LPDDR4 command burst count output – Indicates the number of times a given read or write command is to be repeated by the controller automatically. 0 means one BL16, 1 means two BL16, 63 means 64 BL16
datain_rdy_i	I	1	Data ready input – When asserted, indicates the core is ready to receive the write data. There is 1 or 2 sclk delay between datain_rdy_o to wr_data_i
wr_data_o	O	LPDDR4_DATA_WIDTH	Write data output from Avalon to the memory controller. The write data width is x8 times the memory databus.

read_data_valid_i	I	1	Read data valid output – When asserted, indicates the data on the read_data bus is valid.
rd_data_i	I	LPDDR4_DATA_WIDTH	Read data output from memory controller to the Avalon.

Table 8.13 The Parameter Description of Avalon to LocalBus

Attribute	Selectable Value	Default	Description
Rank Width	1,2	2	The Rank number
Avalon Address Width	27, 28, 29, 30, 31, 32, 33,34	28	The Avalon address width
LocalBus Address Width	26, 27, 28, 29, 30, 31	27	The LPDDR4 address width
Avalon Data Width	32, 64, 128, 256	32	The Avalon data width
LocalBus Data Width	128, 256	128	The LPDDR4 data width
Avalon FIFO Depth	256,512,1024	512	The Avalon FIFO Depth
LocalBus FIFO Depth	128,256,512	128	The LocalBus FIFO Depth
Data Mask Enable	“Enable”, “Disable”	“Enable”	The data mask function enable
databin_rdy_i to wr_data_o delay	1,2	2	Match the LPDDR4 IP

Table 8.14 The address width setting for LocalBus Address Width and Avalon Address Width

Configure	Address Width					
LocalBus Address Width	26	27	28	29	30	31
Avalon Address Width (for LPDDR4 x16 Single Rank)	27	28	29	30	31	--
Avalon Address Width (for LPDDR4 x16 Dual Rank)	--	29	30	31	32	33
Avalon Address Width (for LPDDR4 x32 Single Rank)	28	29	30	31	32	--
Avalon Address Width (for LPDDR4 x32 Dual Rank)	--	30	31	32	33	34

Table 8.14 describes address width setting for LocalBus Address Width and Avalon Address Width.

Table 8.15 The Avalon burst count setting

AVALON_DATA_WIDTH		Avalon burst count			
LocalBus_DATA_WIDTH		32bit	64bit	128bit	256bit
LPDDR4 burst count	128bit	1:8	1:4	1:2	1:1
	256bit	1:16	1:8	1:4	1:2

Table 8.15 describes the Avalon burst count setting of Avalon I/F design.

Below are some usage limitations

- When LPDDR4 Memory data path widths is 16bit, start address aligned with 32n
- When LPDDR4 Memory data path widths is 32bit, start address aligned with 64n
- The Avalon_BurstCount * Avalon_DATA_WIDTH can't be smaller than LocalBus_Burst_Count * LocalBus_DATA_WIDTH*2; For example, Avalon_DATA_WIDTH is 32, LocalBus_DATA_WIDTH is 128, the minimum Avalon_BurstCount is 8.

8.4. Multi-Port Memory Controller reference design

The Multi-Port Memory Controller, also called MPMC, is used when there is more than one user logic need access the Memory Controller, Figure 8.10 shows the MPMC IP GUI. The MPMC supports round robin and order fixed two

arbitration methods, both two arbitrations support up to 9 ports, and each port supports write and read function. [Figure 8.8](#) shows MPMC typical usage.

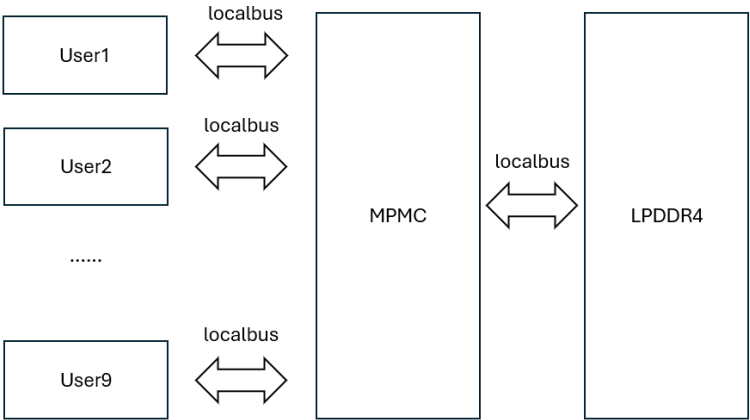


Figure 8.8 MPMC Typical usage

If 9 ports still not enough, multi MPMC also can cascades to support more than 9 ports. [Figure 8.9](#) shows the MPMC cascade usage.

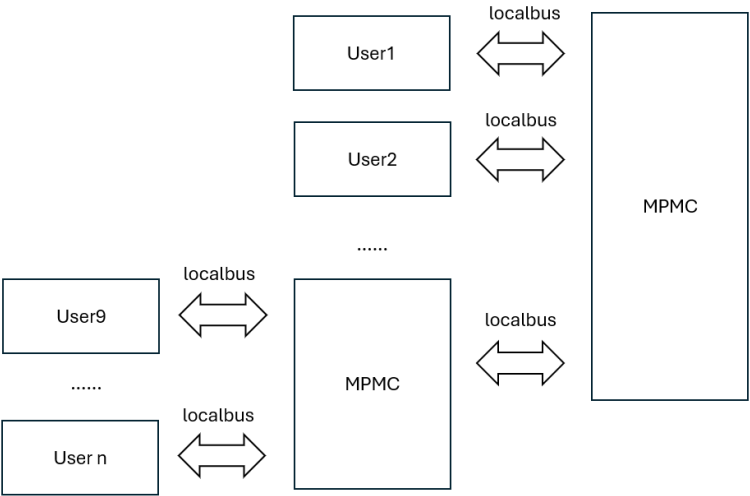


Figure 8.9 MPMC Cascade

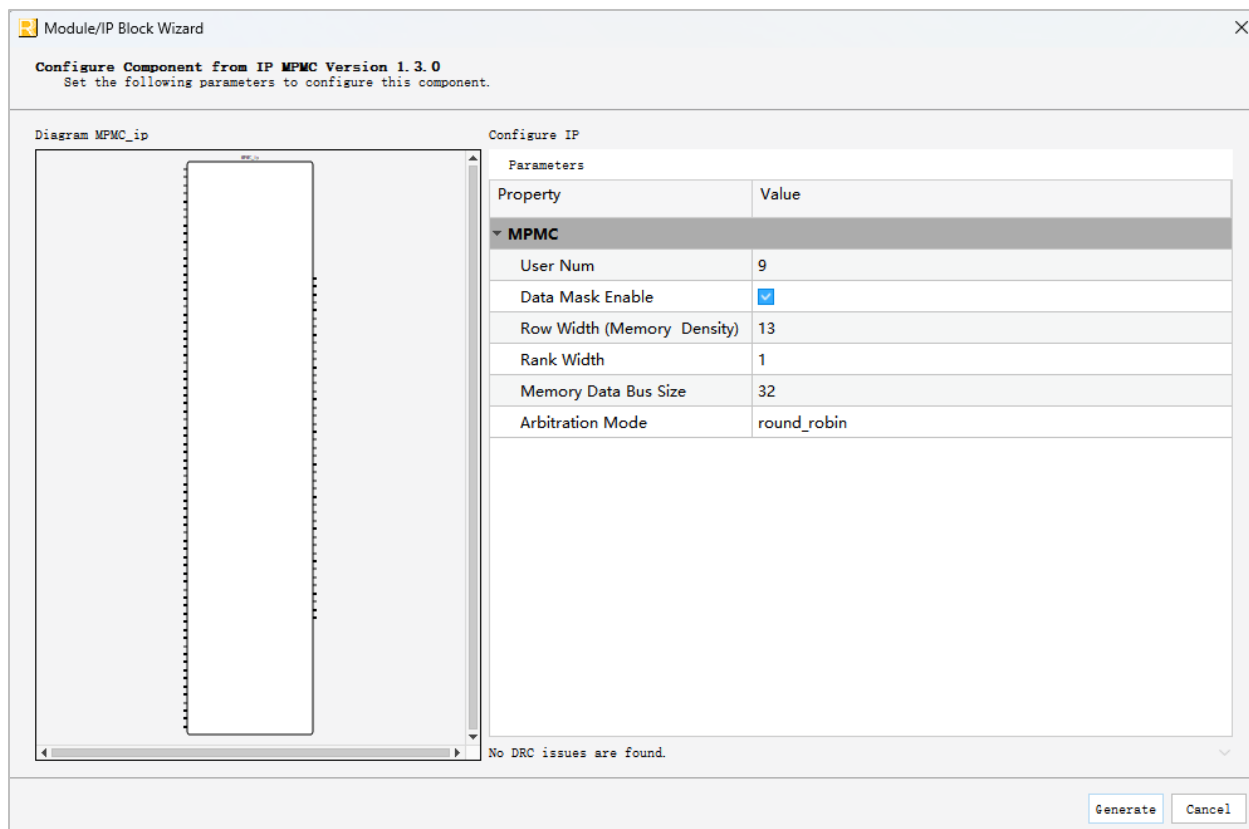


Figure 8.10 MPMC IP GUI

As Table 8.16 shows, the MPMC resource and it depends on port quantity. Due to there is no data buffer so the EBR use is always 0. The efficiency of Round Robin can go up to 81%, the efficiency of Order Fixed can up to 87%. For example, when running at 1066Mbps, the Order Fixed maximum data rate is 927Mbps. MPMC needs extra 4 ddr_sclk to process command, so will add 4 ddr_sclk latency.

Table 8.16 MPMC Resource

	LUTs							
	2 ports	3 ports	4 ports	5 ports	6 ports	7 ports	8 ports	9 ports
Round Robin	80	135	167	249	300	332	416	470
Order Fixed	75	124	155	170	221	251	279	315

When MPMC is set as Order Fixed Mode, user1 always is the highest priority. When higher priority users request to use memory controller, lower priority user' request will be ignored. As shown in Table 8.17 the user' priority is user1>user2>user3>user4>user5>user6>user7>user8. illustrate user' priority order.

Table 8.17 Order Fixed Priority

Users	Priority
User1	★★★★★★★★
User2	★★★★★★★★
User3	★★★★★★★
User4	★★★★★★
User5	★★★★★

User6	★★★★
User7	★★★
User8	★★
User9	★

When MPMC is set as Round Robin. The default priority is same with Order Fixed, but when the user n has been responded, its priority will become lowest. Table 8.18 shows an example to explain round-robin priority, after user3 have been response the priority of all users like below.

If all users always request access to memory controller, in order fixed mode only user1 can succeed in getting control of memory, but in round-robin mode all users can get control of memory sequentially.

Table 8.19 MPMC Parameters and Table 8.20 MPMC Ports describe the parameters and the ports of MPMC, respectively.

Table 8.18 Round Robin Priority

Users	Priority
User1	★★★
User2	★★
User3	★
User4	★★★★★★★★
User5	★★★★★★★★
User6	★★★★★★
User7	★★★★★★
User8	★★★★★
User9	★★★★

Table 8.19 MPMC Parameters

Attribute	Selectable Value	Default	Description
User Num	2-9	2	How many users need access memory controller
Data Mask Enable	Uncheck, Checked	Checked	Write Mask I/F, need keep same with memory controller setting
Row Width (Memory Density)	13,14,15,16,17	13	Memory row width, need keep the same with memory controller setting
Rank Width	1, 2	1	Rank number, 1 is single rank, 2 is dual rank
Memory Data Bus Size	16,24,32	32	Memory data width, need keep same with memory controller setting
Arbitration Mode	round_robin, order_fixed	round-robin	Arbitration algorithm.

Table 8.20 MPMC Ports

Port Name	I/O	Width	Description
System			
rst_n_i	I	1	Asynchronous active low reset.
Connect to Memory Controller			
ddr_sclk_i	I	1	System clock. This is ¼ of the DDR clock frequency and is the main clock of the LPDDR4 Memory Controller IP.
phy_init_success_i	I	1	LPDDR4 initialization and Calibration success, user can write read memory
cmd_valid_o	O	1	LPDDR4 command and address valid output. When asserted, the addr, cmd, ofly_burst_len and cmd_burst_cnt

			inputs are considered valid.
cmd_rdy_i	I	1	LPDDR4 command ready input – When asserted, indicates that the core is ready to accept the next command and the corresponding address.
cmd_wr_1_rd_0_o	O	1 or 2	LPDDR4 command output to the memory controller.
cmd_burst_step_o	O	2	LPDDR4 Burst Step
wr_data_mask_o	O	dq_width	Only available when user enable write mask function. After send write mask command, user needs to input wr_data_mask_i along with wr_data_i.
cmd_addr_o	O	Row Width + 3 + 10 (Single RANK); Row Width+ 1 + 3 + 10 (Dual RANK)	LPDDR4 read or write address output to the memory controller.
cmd_burst_count_o	O	6	LPDDR4 command burst count output – Indicates the number of times a given read or write command is to be repeated by the controller automatically. 0 means one BL16, 1 means two BL16, 63 means 64 BL16 .
datain_rdy_i	I	1	Data ready input – When asserted, indicates the core is ready to receive the write data. There is 1 or 2 sclk delay between datain_rdy_o to wr_data_i
wr_data_o	O	LPDDR4_DATA_WIDTH	Write data output from AXI-4 to the memory controller. The write data width is x8 times the memory databus.
read_data_valid_i	I	1	Read data valid output – When asserted, indicates the data on the read_data bus is valid.
rd_data_i	I	LPDDR4_DATA_WIDTH	Read data output from memory controller to the AXI-4.
Users Interface			
user0_cmd_rdy_o	O	1	The user's I/F, same definition with LPDDR4 Memory Controller
user0_cmd_valid_i	I	1	
user0_cmd_wr_1_rd_0_i	I	1 or 2	
user0_cmd_addr_i	I	Row Width + 3 + 10 (Single RANK); Row Width+ 1 + 3 + 10 (Dual RANK)	
user0_cmd_burst_cnt_i	I	6	
user0_cmd_burst_step_i	I	2	
user0_datain_rdy_o	O	1	
user0_wr_data_i	I	dq_width*8	
user0_wr_data_mask_i	I	dq_width	
user0_rd_data_o	O	dq_width*8	
user0_read_data_valid_o	O	1	

Note: userx_ means user0_ or user1_ or user2_ or user3_ or user4_ or user5_ or user6_ or user7_ or user8_

9.Signal Integrity and PCB Guidelines

9.1. Topology

In PCB design, adopting an appropriate topology structure is crucial for ensuring the stability and performance of external memory systems.

9.1.1. T-Balance

The Memory Controller's PHY can only output a single CA bus for each rank, all LPDDR4 Channels share the CA bus, it means that MC can't set mode register for each channel independently. When designing the PCB it's necessary to ensure VREF of all channels doesn't have too much deviation. Typically, T-balance provides better consistency.

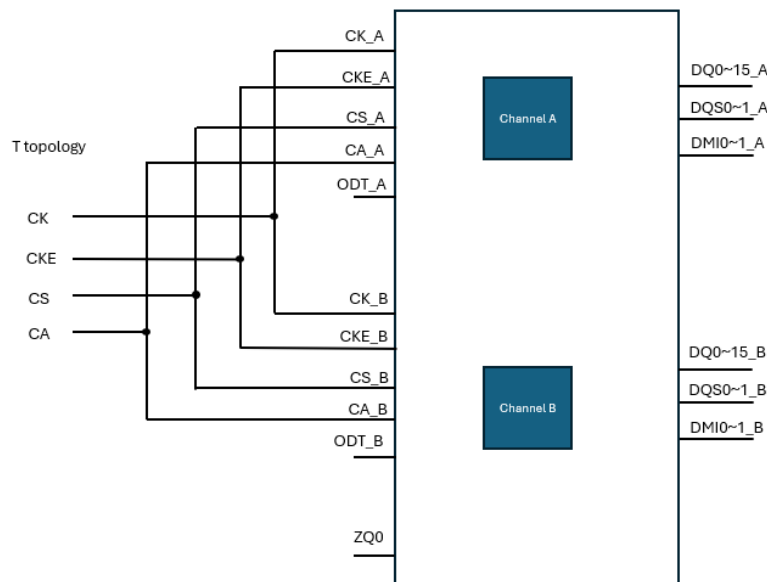


Figure 9.1 T-Balance Topology

9.1.2. Flyby

The "FLY-BY" topology are common memory interface layout that helps reduce signal reflection and crosstalk, and improve signal quality. When the CA channel count exceeds three, a flyby topology usually provides better signal integrity (SI) than a T-balance topology. When designing the PCB, only the last termination needs to be turned on. To ensure successfully write leveling, the CK path delay should be longer than the shortest CA path (first channel CA path).

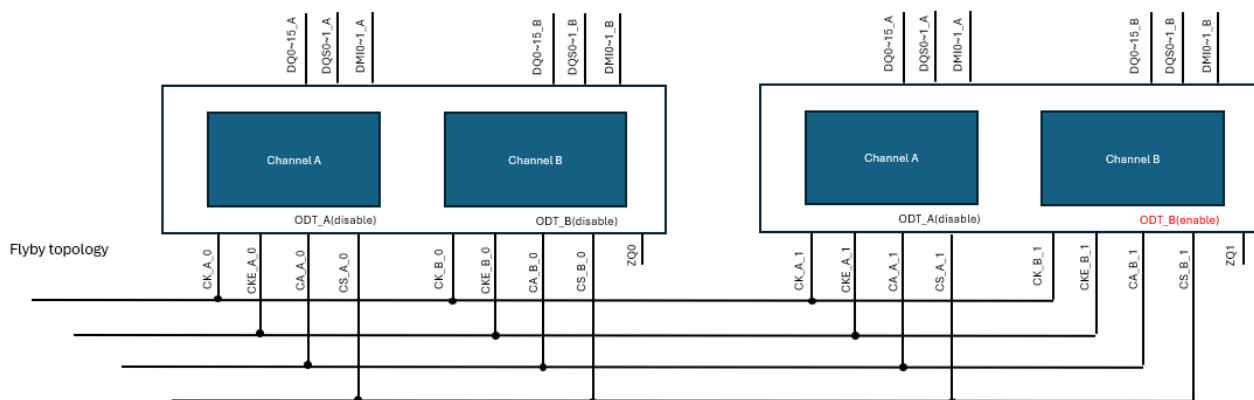


Figure 9.2 Flyby Topology

9.2. Reference Clock

Suggest use dedicated PLL reference clock pin to reduce PLL Jitter. PLL jitter will cause read preamble training fail and limit LPDDR4 Fmax. A 100MHz OSC connected to PLL dedicated IO is recommended. Because Memory connected with HPIO always at bottom of FPGA, the reference clock can't come from top side IO like BANK0, BANK1, BANK7.

9.3. Drive Strength and Termination

Below are some recommending drive strengths and ODT settings which is based on the PCB design described in [CertusPro-NX LPDDR4-1066 Interface Layout Example](#). However, the user maybe needs to fine-tune these settings depending on the variation of PCB design. Usually, the SIPI simulation can help figure out the best settings.

[Table 9.1](#) shows FPGA side settings.

Table 9.1 FPGA Side Settings

Group	IO Type	Slew Rate	Drive Strength	Termination
CA	LVSTL_I	MED	10mA	—
DQ	LVSTL_I	FAST	10mA	60 ohm
DQS	LVSTLD_I	FAST	10mA	60 ohm

Based on the test experience, the [Table 9.2](#) is presented to show the default settings on the LPDDR4 side.

Table 9.2 LPDDR4 Side Settings

Data rate	Items	Value
<=1066Mbps	CA ODT	RZQ/1
	DQ ODT	RZQ/1
	PDDS	RZQ/6
	PU-CAL	VDDQ/3
>1066Mbps	CA ODT	RZQ/1
	DQ ODT	RZQ/4
	PDDS	RZQ/6
	PU-CAL	VDDQ/3

9.4. CertusPro-NX LPDDR4-1066 Interface Layout Example

This section described the LPDDR4 layout guidelines to meet JEDEC LPDDR4 spec

Table 9.3 PCB Layout general guidance

Group	Parameter	Condition
Data Bus	Trace Length	< 2 inches*
	Impedance	40 to 50 ohm
	Connection	Point to point
	Routing Type	Microstrip/Stripline
	Length matching	<+-40mil reference to DQS trace
	DQS P/N matching	<+-4mil
	DQS impedance	80 to 100 ohm
Command/Address, Chip Selection, Clock	Trace Length	<2 inches*
	Impedance	40 to 50 ohm
	Connection	T-point or Flyby at memory for dual Channels
	Routing type	Microstrip/Stripline
	Length matching	<+-40mil reference to clock trace
	Clock P/N matching	<+-4mil
	Clock impedance	100 ohm Differential

*: IP Timing need be checked if trace length is longer than the spec.

Three inner signal layers (L03/L05/L08) in 10-layer FR4 Stackup are mainly used to route the LPDDR4 interface signals between the CertusPro-NX memory controller(U1) and a Micron commercial Mobile LPDDR4 4Gbit memory(U7), due to surface layers are stuffed with landing pads and decoupling capacitors.

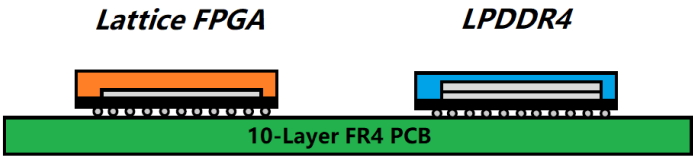


Figure 9.3 Lattice LPDDR4 controller subsystem on PCB

The stack structure was showed as below with impedance control requirements. For LPDDR4 signals, this board required 50 Ohm for single end signals (DQ and CA) and 100 Ohm for differential signals (DQS and Clock pairs)

	Subclass Name	Type	Material	Thickness (MIL)
TOP	TOP	CONDUCTOR	COPPER	1.4
DL_002		DIELECTRIC	FR-4	3.9
L02-GND	L02-GND	PLANE	COPPER	1.2
DL_004		DIELECTRIC	FR-4	5.12
L03-SIG1	L03-SIG1	CONDUCTOR	COPPER	1.2
DL_006		DIELECTRIC	FR-4	6
L04-POW1	L04-POW1	PLANE	COPPER	1.2
DL_008		DIELECTRIC	FR-4	5.12
L05-SIG2	L05-SIG2	CONDUCTOR	COPPER	1.2
DL_010		DIELECTRIC	FR-4	6
L06-GND	L06-GND	PLANE	COPPER	1.2
DL_012		DIELECTRIC	FR-4	5.12
L07-POW2	L07-POW2	PLANE	COPPER	1.2
DL_014		DIELECTRIC	FR-4	6
L08-SIG3	L08-SIG3	CONDUCTOR	COPPER	1.2
DL_016		DIELECTRIC	FR-4	5.12
L09-GND	L09-GND	PLANE	COPPER	1.2
DL_018		DIELECTRIC	FR-4	3.9
BOTTOM	BOTTOM	CONDUCTOR	COPPER	1.4

Figure 9.4 Reference PCB Stack information

Signal layer L03 is used to route DQ[0..7] of channel A and signal layer L05 is used to route DQ[8..15] of channel A, as shown in the following picture:

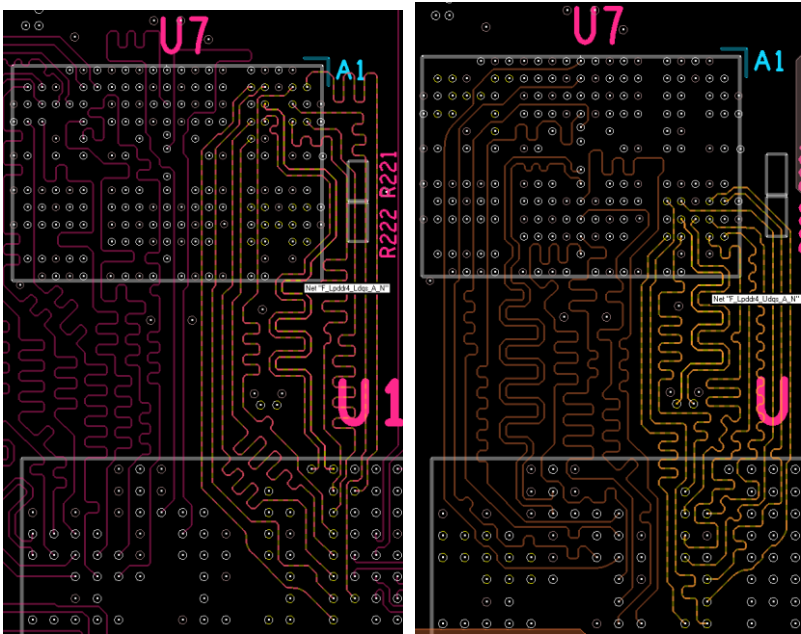


Figure 9.5 LPDDR4 DQS group A routing Example

Table 9.4 DQS group A trace information of example PCB

Signal name	Layer	Length	Trace Width	Impedance
LDQS_A_P	L03_SIG1	1178.34 MIL	3.5/8/3.5mil	100±10 ohm
LDQS_A_N	L03_SIG1	1178.59 MIL		
DQ0_A	L03_SIG1	1178.44 MIL	3.9 mil	50±5 ohm
DQ1_A	L03_SIG1	1179.07 MIL	3.9 mil	50±5 ohm
DQ2_A	L03_SIG1	1178.56 MIL	3.9 mil	50±5 ohm
DQ3_A	L03_SIG1	1179.58 MIL	3.9 mil	50±5 ohm
DQ4_A	L03_SIG1	1180.06 MIL	3.9 mil	50±5 ohm
DQ5_A	L03_SIG1	1177.47 MIL	3.9 mil	50±5 ohm
DQ6_A	L03_SIG1	1175.97 MIL	3.9 mil	50±5 ohm
DQ7_A	L03_SIG1	1177.47 MIL	3.9 mil	50±5 ohm
LDM_A	L03_SIG1	1178.45 MIL	3.9 mil	50±5 ohm
UDQS_A_P	L05_SIG2	973.45 MIL	3.5/8/3.5mil	100±10 ohm
UDQS_A_N	L05_SIG2	973.46 MIL		
DQ8_A	L05_SIG2	972.22 MIL	3.9 mil	50±5 ohm
DQ9_A	L05_SIG2	973.05 MIL	3.9 mil	50±5 ohm
DQ10_A	L05_SIG2	974.67 MIL	3.9 mil	50±5 ohm
DQ11_A	L05_SIG2	974.62 MIL	3.9 mil	50±5 ohm
DQ12_A	L05_SIG2	974.93 MIL	3.9 mil	50±5 ohm
DQ13_A	L05_SIG2	972.64 MIL	3.9 mil	50±5 ohm
DQ14_A	L05_SIG2	975.49 MIL	3.9 mil	50±5 ohm
DQ15_A	L05_SIG2	971.52 MIL	3.9 mil	50±5 ohm
UDM_A	L05_SIG2	973.19 MIL	3.9 mil	50±5 ohm

Signal layer L03 is used to route DQ[7:0] of channel B and signal layer L08 is used to route DQ[15:8] of channel B, as shown in the following picture:

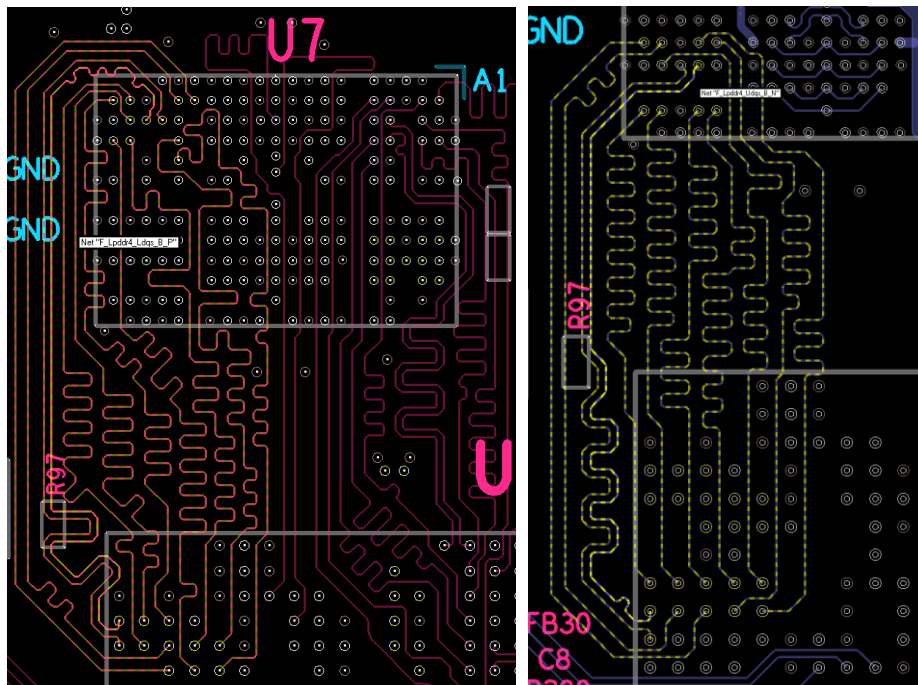


Figure 9.6 LPDDR4 DQS group B routing Example

Table 9.5 DQS group B trace information of example PCB

Signal name	Layer	Length	Trace Width	Impedance
LDQS_B_P	L03_SIG1	1489.71 MIL	3.5/8/3.5mil	100±10 ohm
LDQS_B_N	L03_SIG1	1489.17 MIL		
DQ0_B	L03_SIG1	1489.58 MIL	3.9 mil	50±5 ohm
DQ1_B	L03_SIG1	1489.86 MIL	3.9 mil	50±5 ohm
DQ2_B	L03_SIG1	1488.52 MIL	3.9 mil	50±5 ohm
DQ3_B	L03_SIG1	1492.05 MIL	3.9 mil	50±5 ohm
DQ4_B	L03_SIG1	1489.69 MIL	3.9 mil	50±5 ohm
DQ5_B	L03_SIG1	1488.02 MIL	3.9 mil	50±5 ohm
DQ6_B	L03_SIG1	1490.2 MIL	3.9 mil	50±5 ohm
DQ7_B	L03_SIG1	1491.17 MIL	3.9 mil	50±5 ohm
LDM_B	L03_SIG1	1489.35 MIL	3.9 mil	50±5 ohm
UDQS_B_P	L08_SIG3	1200.54 MIL	3.5/8/3.5mil	100±10 ohm
UDQS_B_N	L08_SIG3	1199.69 MIL		
DQ8_B	L08_SIG3	1199.6 MIL	3.9 mil	50±5 ohm
DQ9_B	L08_SIG3	1200.11 MIL	3.9 mil	50±5 ohm
DQ10_B	L08_SIG3	1199.4 MIL	3.9 mil	50±5 ohm
DQ11_B	L08_SIG3	1201.09 MIL	3.9 mil	50±5 ohm
DQ12_B	L08_SIG3	1202.38 MIL	3.9 mil	50±5 ohm
DQ13_B	L08_SIG3	1201.3 MIL	3.9 mil	50±5 ohm
DQ14_B	L08_SIG3	1201.58 MIL	3.9 mil	50±5 ohm
DQ15_B	L08_SIG3	1201.83 MIL	3.9 mil	50±5 ohm
UDM_B	L08_SIG3	1200.11 MIL	3.9 mil	50±5 ohm

3 layers are used to route CA bus shown in the following picture, all single rank CA signals were routed in L03 and dual rank supported signals were added in L05.

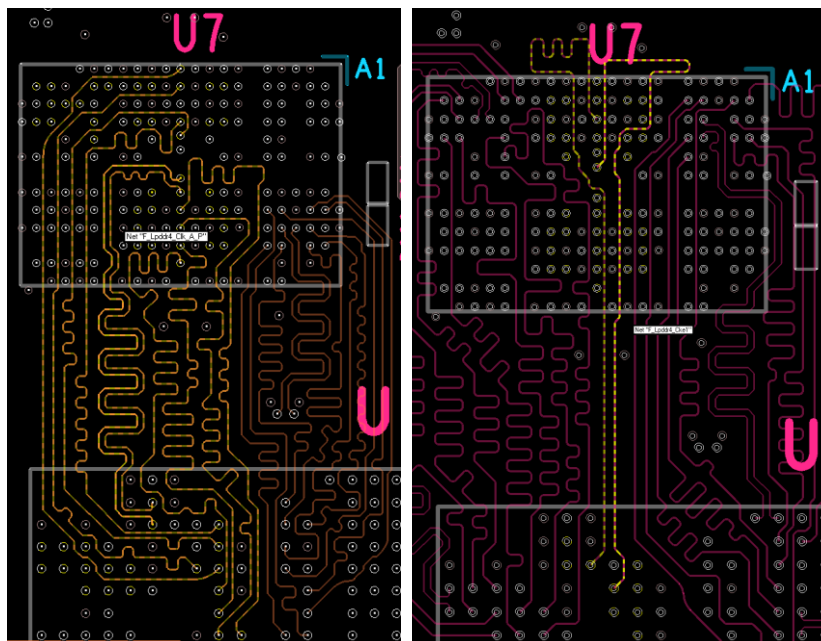


Figure 9.7 LPDDR4 CA group routing Example with Dual Rank support

The shared bus for dual channel support was implemented with T-point which were split in L08, the split path should be less than 100 mil post vias for each node.

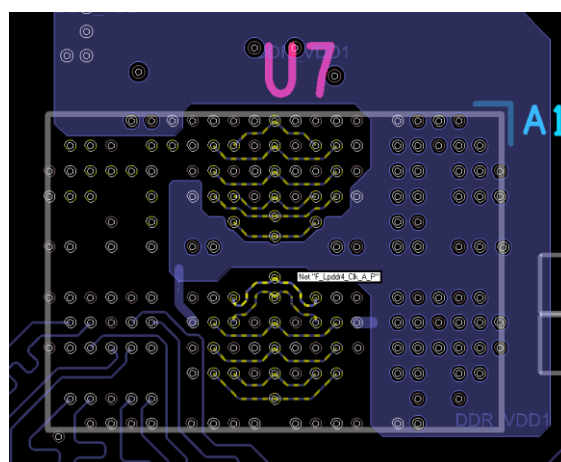


Figure 9.8 LPDDR4 CA group split Example with T-type topology

Table 9.6 CA group trace information of example PCB

Signal name	Share Layer	Split Layer	Total Length	Trace Width	Impedance
CA0	L05_SIG2	L08_SIG3	1555.54 MIL	3.9 mil	50±5 ohm
CA1	L05_SIG2	L08_SIG3	1545.06 MIL	3.9 mil	50±5 ohm
CA2	L05_SIG2	L08_SIG3	1554.37 MIL	3.9 mil	50±5 ohm
CA3	L05_SIG2	L08_SIG3	1562.8 MIL	3.9 mil	50±5 ohm
CA4	L05_SIG2	L08_SIG3	1553.9 MIL	3.9 mil	50±5 ohm
CA5	L05_SIG2	L08_SIG3	1526.85 MIL	3.9 mil	50±5 ohm
CKE0	L05_SIG2	L08_SIG3	1531.24 MIL	3.9 mil	50±5 ohm

CS0	L05_SIG2	L08_SIG3	1568.63 MIL	3.9 mil	50±5 ohm
CKE1	L03_SIG1	L08_SIG3	1521.35 MIL	3.9 mil	50±5 ohm
CS1	L03_SIG1	L08_SIG3	1568.79 MIL	3.9 mil	50±5 ohm
CLK_P	L05_SIG2	L08_SIG3	1577.01 MIL	3.5/8/3.5mil	100±10 ohm
CLK_N	L05_SIG2	L08_SIG3	1578.14 MIL		

Based on Hyperlynx simulation results, The following parameters are suggested to be implemented for the CertusPro-NX memory controller IP.

Table 9.7 LPDDR4 IP configuration parameters based on example PCB

Configuration Item	Value
LPDDR4 Controller Slew rate for DQ/DQS/DM	Fast
LPDDR4 Controller Drive Strength for DQ/DQS/DM	10mA
LPDDR4 Controller Slew Rate for CA/CS/CK	Fast
LPDDR4 Controller Drive Strength for CA/CS/CK	10mA
LPDDR4 Memory-Receiver for CA/CS/CK (Write & Read)	ODT-240ohm
LPDDR4 Memory -Receiver for DQ/DM/DQS (Write)	ODT-240ohm
LPDDR4 Memory -Driver for DQ/DM/DQS (Read)	PD40-ODT40-VOH30
LPDDR4 Controller Receiver for DQ/DM/DQS (Read)	ODT-40ohm

References

1. JESD209-4 Low Power Double Data Rate4 (LPDDR4)
2. DFI4.0 Specification

Technical Support Assistance

Submit a technical support case through www.latticesemi.com/techsupport.

Revision History

IP Revision

Section	Change Summary
1-7	2024-06-26: First release, 1.0.0 version
1-7	2024-07-26: 1.1.0 version 1) Add write mask function
All	2024-08-01: 1.2.0 version 1) Add Self-Refresh function 2) Add Interrupt Register 3) Add Status Registers 4) Add PCB Guideline 5) Update Stream Interface 6) Add AXI-4 Interface
All	2024-12-31: 1.3.0 version 1) Add 750MHz(1.5Gbps) support 2) Add Dual Rank support 3) Remove WL/RL setting 4) Change Stream Interface to Virtual FIFO
All	2025-12-31: 1.4.0 version 1) Add 700MHz(1.4Gbps) support 2) Support user define burst step 3) Evaluation project and test bench is updated
All	2026-06-22: 1.4.1 version 1) Add chapter 8.1.1 2) Add chapter 8.1.2 3) Table 3.2, 5.1, 5.2, 5.3 updated



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